# Information Provision with Intertemporal Externalities*

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**Abstract**

I analyse a dynamic model of persuasion in which a long-lived Sender provides information about an evolving state to a sequence of short-lived Receivers who choose between two actions, high and low. In each period, the Sender has a weak ordinal preference for the high action over the low and the good state over the bad, whereas each Receiver wishes to match action to state. Importantly, each Receiver's action affects the state the next one faces, creating intertemporal externalities: the high action may either improve or worsen the future state. When the high action improves the future state, the Sender's ordinal preferences alone ensure that the optimal policy maximises the probability of the high action each period. When the high action worsens the future state, the optimal policy depends on the cardinal features of her preferences: how she values current actions relative to the future state. She may then maximise the probability of the high action or the low action, fully reveal the state, or use an intermediate policy that leaves the Receiver uncertain. Beyond *what* the Sender discloses, the optimal policy also determines *when* she discloses, and whether disclosure eventually stops.

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# 1 Introduction

In many economic environments, today's actions affect not only today's outcomes but also the opportunities and payoffs available in the future. The quality of natural resources depends on current conservation decisions;[1] investment affects future firm profitability;[2] parents' investment in education affects their children's future earnings;[3] and current interpersonal contact affects the future spread of contagious diseases.[4] A common feature of these environments is that current actions generate *intertemporal externalities*: they impose benefits or costs on future decision-makers by changing the state of the world they will inherit.

Consider fisheries. Fish stocks are a common-property resource and are therefore vulnerable to over-exploitation, generating the classic "tragedy of the commons" (Gordon, 1954; Hardin, 1968). Because current extraction affects the future stock of fish, the current use of the resource creates an intertemporal trade-off: greater extraction raises current output but can reduce future availability. Responses to this problem include assigning property rights (Scott, 1955) or governing the resource through community institutions (Ostrom, 1990). In international waters, however, such solutions are hard to implement, and enforcement is comparatively limited. As a result, regional fisheries advisory bodies play a central role there, providing the scientific information on which fishing decisions are based. The International Council for the Exploration of the Sea (ICES), for example, collects data on fish stocks in the North Atlantic and adjacent seas and publishes its assessments. Fishing fleets observe this information and choose their level of fishing effort—how many ships to send out and how many nets to cast—to maximise their profit: high effort pays when the stock is abundant, but not when it is depleted. Thus, an expert's information disclosure can affect not only the decision taken today, but also the state of the resource that future decision-makers will face.

This observation motivates the following question: *how should an expert provide information when the decision she induces today affects the state of the world tomorrow?* I study this question

[1] This is the motivation underlying the notion of sustainable development; see World Commission on Environment and Development (1987).

[2] See, for example, Harris and Raviv (1991) and Myers (2001).

[3] See Mincer (1958), Schultz (1961), Becker (1962), and Becker and Tomes (1979).

[4] See Kermack and McKendrick (1927); for economic analyses of policies that reduce interpersonal contact, see, for example, Geoffard and Philipson (1996) and Adda (2016).

in a stylised infinite-horizon model of information provision. A long-lived Sender with commitment power interacts with a sequence of short-lived Receivers. At the beginning of each period, a payoff-relevant state is realised, which neither the Sender nor the Receiver observes. Instead, both share a common prior and observe the public history of signals and actions. In every period, the Sender chooses and commits to a Blackwell experiment. The resulting signal is publicly observed, and both players update their beliefs to a posterior, after which the Receiver takes a myopically optimal action. The current state and the Receiver's action then determine the transition probabilities governing the state in the next period. The Sender maximises her discounted payoff over the infinite horizon.

I focus on a binary state, which can be interpreted as a good or bad level of the resource, and a binary action, which can be interpreted as a high or low level of effort. The Sender has a weak ordinal preference for the good state over the bad and the high action over the low. This structure captures the trade-off faced by a resource-management body that values current production and employment but also cares about the long-run viability of the resource.[5] The Receiver, by contrast, wants to match his action to the current state: he takes the high action when the state is good and the low action when the state is bad.

The resulting problem has a simple interpretation. Within a period, the Sender would like to use information to maximise the probability of the high action, subject to a credibility constraint. Across periods, however, she would like to steer the state towards the good state. When there are *positive intertemporal externalities*, i.e. the high action improves the distribution of future states, these two objectives are aligned. When there are *negative intertemporal externalities*, i.e. the high action worsens the distribution of future states, they conflict.

I show that this distinction sharply determines the optimal policy. The *greedy policy* applies the optimal static information structure in every period, thereby maximising the probability of the high action given the prevailing prior in each period. Under positive intertemporal externalities, the Sender's weak ordinal preferences for the high action and the good state are sufficient for the

[5]The ICES, for example, states that it is "committed to better understanding marine ecosystems and securing the benefits that people derive from them." The body "generate[s] and share[s] the data, knowledge, and advice needed to meet current and emerging conservation, management, and sustainability goals" (International Council for the Exploration of the Sea, 2026).

greedy policy to be optimal (Theorem 1). The reason is that positive externalities align the Sender's static and dynamic objectives, so there is no reason to sacrifice current persuasion for future state management.

Under negative intertemporal externalities, the Sender faces a trade-off between the current payoff from inducing the high action and the future benefit of improving the state distribution. Since the high action worsens the distribution of future states, inducing it more often comes at a dynamic cost. Therefore, the optimal policy depends on how the Sender's cardinal preferences weigh these competing objectives.

Theorem 2 characterises the optimal policy. When the Sender's preference for the high action is sufficiently strong relative to her preference for the good state, the greedy policy is optimal. This includes the case in which the Sender has *cardinal state-independent preferences:* she receives a positive payoff if and only if the Receiver takes the high action, and zero otherwise, regardless of the state. Conversely, when the Sender's preference for the good state is sufficiently strong, the *judicious policy*, which maximises the probability of the low action every period given the prevailing prior, is optimal. When the Sender's preferences for both the high action and the good state are sufficiently weak, *full disclosure* is optimal: the benefits of inducing actions that conflict with the Receiver's interests are too small to outweigh their costs.

When the Sender's preferences for both the high action and the good state are sufficiently strong, the optimal policy involves *interior pooling.* Unlike full disclosure, which reveals the state to the Receiver, interior pooling leaves him uncertain after either signal: he assigns positive probability to both states. In other words, the experiment generates two posterior beliefs, both strictly between zero and one. One induces the high action and the other the low action, so the Receiver may take the high action in the bad state or the low action in the good state.

It is this uncertainty that allows the Sender to strike a different balance from either of the two action-oriented policies. Compared to the greedy policy, the Sender gives up some chance of inducing the high action to induce the low action more often and improve the future state distribution. Compared to the judicious policy, she retains a chance of inducing the high action and securing its current payoff benefit. Interior pooling thus allows the Sender to pursue both

objectives without focusing exclusively on either action or fully revealing the state.

The analysis also yields comparative-statics insights into how patience and state transitions shape information provision (Proposition 3). They reveal that greater patience does not simply make the Sender more transparent. A more patient Sender becomes more willing to reveal the bad state, where inducing the high action harms future prospects, but more willing to withhold information in the good state, where inducing the low action improves them. Holding her preferences fixed, changes in the transition probabilities have similarly state-specific effects.

Finally, I show that the optimal policy determines not only *what* information the Sender reveals, but also *when* she reveals it. This is important in applications such as fisheries management, where the fishing effort induced by one stock assessment changes the resource inherited by future fleets, thereby shaping when further information is needed. Under every policy other than full disclosure, the Sender communicates only when the current belief lies within a certain range. At each disclosure, her experiment generates one of two posterior beliefs: one induces the high action and the other the low action. The posterior and the action taken determine the next period's prior, after which beliefs evolve predictably if the Sender remains silent. Thus, after the first disclosure, beliefs follow one of two paths, depending on which posterior the experiment induces at each disclosure, and these paths determine when the Sender next communicates (Lemma 2).

Whether disclosure eventually stops depends on whether an action can be sustained without further information. Consider the greedy policy and suppose a signal is realised that induces the high action. If taking that action leaves the next Receiver willing to choose the high action without new information, disclosure is needed in only finitely many periods (Proposition 4). Otherwise, the Sender must disclose again immediately after inducing the high action, and disclosure recurs indefinitely (Proposition 5). The judicious policy exhibits the reverse pattern. These results show that the content of a message determines when the next one is needed.

The rest of the paper is organised as follows. Next, I review the related literature. Section 2 introduces a two-state, two-action, infinite-horizon model of information provision with intertemporal externalities. Section 3 characterises the static benchmark, derives the Sender's Bellman equation and the properties of optimal policies, and solves two dynamic benchmarks. Sections 4 and

5 characterise the optimal policy under positive and negative externalities, respectively. Section 6 describes the belief dynamics and disclosure patterns that the optimal policies generate. Section 7 concludes. All proofs are in the appendices.

### 1.1 Related Literature

This paper belongs to the literature on dynamic information provision, which builds on the insights in Aumann and Maschler (1995) and Kamenica and Gentzkow (2011). The closest papers to the model analysed in this paper are Renault, Solan, and Vieille (2017), Ely (2017), and Ashkenazi-Golan et al. (2023).[6] In their settings, a single long-lived Sender communicates with a Receiver who behaves myopically, as here. There are *exogenous* (Markovian) state transitions, where the Sender cares about the evolution of the state but cannot influence it through the Receiver's action. In contrast, here, information affects current actions, and those actions in turn affect the future state.

Renault, Solan, and Vieille (2017) analyse a model with a finite state space and a binary action space, in which the Sender has cardinal state-independent preferences: in every period, she only wants to maximise the probability that a particular action is taken. The authors identify a sufficient condition under which the greedy policy is also optimal in the dynamic game. Ashkenazi-Golan et al. (2023) analyse a model with a binary state space and a finite action space, in which the Sender also has cardinal state-independent preferences. In the basic model of Ely (2017), the state and action spaces are binary, with one of the states being absorbing; the Sender again has cardinal state-independent preferences. In all three papers, if the state and action spaces are binary, then the greedy policy is optimal in the dynamic game.[7] I show that, under positive externalities, a weak ordinal preference for the high action and the good state is sufficient for the greedy policy to be optimal. On the other hand, under negative externalities, the optimal policy depends on cardinal features of the Sender's preferences, and need not be the greedy policy.

[6] Lehrer and Shaiderman (2026) also study persuasion about a state that follows an exogenous Markov chain, but ask a different question: for general finite state and action spaces and state-dependent payoffs, they characterise when a patient Sender attains the static value at the stationary belief.

[7] If the state space or the action space has more than two elements, the greedy policy need not be optimal in the dynamic game, even with exogenous transitions and cardinal state-independent Sender preferences; see Renault, Solan, and Vieille (2017) and Ashkenazi-Golan et al. (2023).

Other papers study dynamic information provision to long-lived Receivers, where information also acts as an incentive over time. Ely and Szydlowski (2020) show how a principal who knows the difficulty of a task can use information over time to keep an agent working on it. Ball (2023) analyses a model in which both players are long-lived and the state evolves exogenously. Sugaya and Wolitzky (2026) analyse a model with a single Sender and multiple Receivers, all long-lived. In the baseline model, the states are i.i.d., but the authors consider exogenous Markov transitions in an extension. In all three papers, the Sender commits at the outset to a policy for the whole game. In Ball (2023) and Sugaya and Wolitzky (2026), for example, the optimal policy is implemented by threatening to cut off all future information after any deviation. Here, because each Receiver is short-lived, such threats have no force, and the Sender's commitment power over time is equivalent to commitment within a period.[8]

A few papers also study a Sender who faces several short-lived Receivers in turn, but the Receivers are linked through different channels. In Kremer, Mansour, and Perry (2014), Receivers choose between two actions with unknown rewards, and each choice reveals its reward to the Sender. Thus, Receivers' actions improve future payoffs through learning. In the sequential procedure in Khantadze, Kremer, and Skrzypacz (2025), each Receiver cares only about his own state, but the states are correlated, so information given to one Receiver constrains the persuasion of the others. Here, Receivers are linked through the state: each Receiver's action changes the state that the next one faces.

Finally, from a modelling perspective, this paper can also be viewed as a stochastic game (Shapley, 1953; Solan & Vieille, 2015) in which one player's action affects the transition probabilities as in Renault (2012). There, however, the controlling player is also informed of the state, whereas here the Receiver controls the transitions and holds no private information.

[8] Without commitment, Renault, Solan, and Vieille (2013) characterise the limit equilibrium payoffs of a sender–receiver game in which the state follows a Markov chain observed only by the sender.

## 2 Model

In this section, I introduce a two-state, two-action, infinite-horizon model of information provision with intertemporal externalities. Time, indexed by $t \in \mathbb{N}$, is discrete.

**Agents and payoffs.** A long-lived Sender (she) interacts with a sequence of short-lived Receivers (he). Their stage game payoffs depend on the state of the world $\omega_t \in \Omega = \{G(\text{ood}), B(\text{ad})\}$ and the Receiver's action $a_t \in \mathcal{A} = \{h(\text{igh}), \ell(\text{ow})\}$ in that period, and are given as follows:

| $u_S(\omega_t, a_t)$ | $\omega_t = G$ | $\omega_t = B$ |
|---|---|---|
| $a_t = h$ | 1 | $x$ |
| $a_t = \ell$ | $y$ | 0 |

| $u_R(\omega_t, a_t)$ | $\omega_t = G$ | $\omega_t = B$ |
|---|---|---|
| $a_t = h$ | 1 | $-1$ |
| $a_t = \ell$ | 0 | 0 |

where $x \in [0, 1]$ and $y \in [0, 1]$.

Clearly, within a period, both players weakly prefer state $G$ to state $B$; the Sender's *ordinal* preference is strict if $x < 1$ and $y > 0$. The Sender also weakly prefers action $h$ to action $\ell$ (strict *ordinal* preference if $x > 0$ and $y < 1$). The Sender has *cardinal state-independent* stage game preferences if and only if $x = 1$ and $y = 0$. Likewise, the Sender has *cardinal action-independent* stage game preferences if and only if $x = 0$ and $y = 1$. Meanwhile, the Receiver wants to match action to state: he strictly prefers taking action $h$ in state $G$ and action $\ell$ in state $B$.

Since the Receiver is short-lived, he chooses an action that maximises his stage game payoff. On the other hand, the Sender maximises her long-run expected discounted payoff:

$$(1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} \mathbb{E}[u_S(\omega_t, a_t)],$$

where $\delta \in [0, 1)$ is the Sender's discount factor.

**Intertemporal externalities.** The key departure from previous models of dynamic information provision is that the distribution of next period's state depends jointly on the current state and the Receiver's current action. Thus, the state evolves *endogenously* over time. Formally, transition probabilities are given by $\mathbb{P}(\omega_{t+1} \mid \omega_t, a_t)$, such that $\sum_{\omega_{t+1} \in \Omega} \mathbb{P}(\omega_{t+1} \mid \omega_t, a_t) = 1$. Since the state

and action spaces are binary, the transition probabilities can be described by the following matrix:

| $\mathbb{P}(\omega_{t+1} = G \mid \omega_t, a_t)$ | $\omega_t = G$ | $\omega_t = B$ |
|---|---|---|
| $a_t = h$ | $p^h_G$ | $p^h_B$ |
| $a_t = \ell$ | $p^\ell_G$ | $p^\ell_B$ |

(2.1)

I use the term *intertemporal externality* to describe the effect of a current action on future states and hence on future payoffs. Since Receivers are short-lived, these effects are external to the Receiver choosing the current action. In particular, the Receiver's period-$t$ action may impose a benefit or a cost on future periods by changing the likelihood that the state is good in period $t+1$:

**Definition 1** (Intertemporal externalities)**.** The high action imposes a **positive intertemporal externality** if $p^h_\omega > p^\ell_\omega \ \forall \omega \in \Omega$, and a **negative intertemporal externality** if $p^h_\omega < p^\ell_\omega \ \forall \omega \in \Omega$. State transitions are **exogenous** if $p^h_\omega = p^\ell_\omega \ \forall \omega \in \Omega$.

Therefore, under positive (negative) intertemporal externalities, conditional on either state, choosing the high action makes the good state more (less) likely in the following period than choosing the low action. In what follows, for the sake of brevity, I omit the term "intertemporal".

**Timing.** At the beginning of a period, conditional on the public history of past signals and actions, both players share a common prior $p_t \coloneqq \mathbb{P}(\omega_t = G) \in [0, 1]$. Every period, the Sender publicly chooses an *information structure* (that is, a Blackwell experiment; Blackwell, 1951), whose signal realisation is publicly observed. A *policy* specifies the experiment that the Sender chooses in each period as a function of the public history.

Formally, an information structure is a pair $(\Sigma, \sigma_t)$, where $\Sigma$ is a set of signals, and $\{\sigma_t(\cdot \mid \omega)\}_{\omega \in \Omega}$ is a set of probability distributions over signals, conditional on the true state $\omega$. The Receiver observes the chosen information structure and both players observe the signal realisation $\varsigma_t \in \Sigma$, update their belief about the state using Bayes' rule to $q_t = \mathbb{P}(\omega_t = G \mid \varsigma_t)$, and the Receiver chooses an action that maximises his stage game payoffs.[9] Based on the action chosen by the Receiver, the state evolves for the next period (see Figure 1).

[9] I assume that the Receiver breaks ties in the Sender's favour taking into account her continuation payoff, so that the Sender's interim expected payoffs are upper semi-continuous. This is needed to ensure existence of the Sender's optimum.

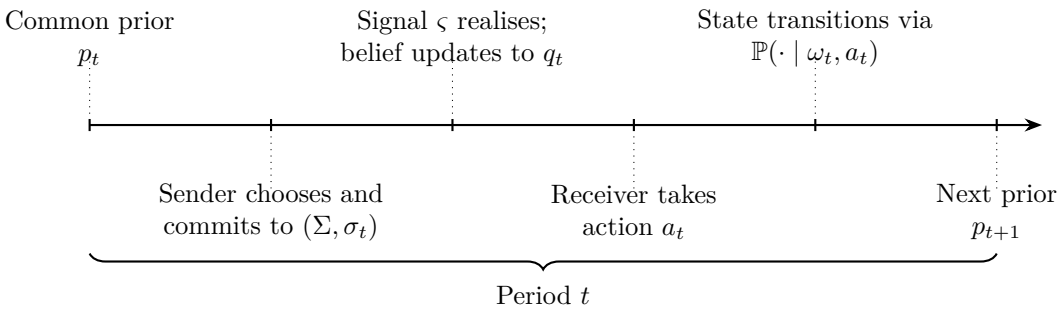



Figure 1: Timeline of events within a single period

**Information.** Both players know the stage game payoffs $u_S$, $u_R$ and the transition probabilities $\{p^a_\omega\}$. Realised stage payoffs are not observed by either player (equivalently, they accrue only at the end of the game), so all learning about the state is through the Sender's signals. In every period $t$, both players observe the entire public history of past signals and past actions. Neither player ever directly observes the realised state $\omega_t$, so the Sender has no private information in this setup.[10] She updates her belief in the same manner as the Receiver: after observing the signal realisation and after the Receiver takes an action. Thus, in each period $t$, conditional on the public history and the current signal realisation, both players share the same posterior belief $q_t$ and the same prior for the next period $p_{t+1}$.

# 3 Preliminaries

In this section, I first characterise the Sender's static optimum. I then derive the Sender's Bellman equation and the properties of optimal policies, and solve two dynamic benchmarks.

## 3.1 Static Benchmark: The Greedy Policy

In any period, the problem facing the Receiver is simple. Given his period-$t$ posterior belief $q_t$, the Receiver gets an expected payoff of $q_t \cdot 1 + (1 - q_t) \cdot (-1)$ if he chooses action $h$, and a payoff of 0 if

[10] As mentioned in Renault, Solan, and Vieille (2017), an alternative interpretation would be that the Sender observes the sequence of states and chooses and commits to a policy *ex-ante*, i.e. prior to the beginning of the game. This is because the Receiver is short-lived, so the Sender's problem reduces to a single-agent stochastic dynamic optimisation problem in which she is constrained by the Receiver's best-response function. This has a recursive formulation: see Section 3.2.2 and equation (3.4) below. In this case, a policy for the Sender would map histories (past states and actions and current state) into distributions over signals.

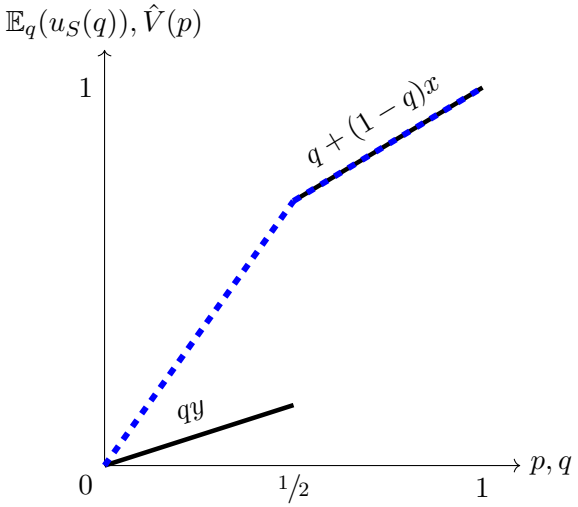



Figure 2: Sender's interim (black) and optimal (blue) expected payoff under the greedy policy

he chooses action $\ell$. Thus, the Receiver's best response is to choose $a_t = h$ if $q_t \geq 1/2$ (at $q_t = 1/2$, he breaks his indifference in the Sender's favour, which in the static game means taking the high action) and $a_t = \ell$ otherwise. Given this best-response function, the Sender's interim expected payoffs are:

$$\mathbb{E}_{q_t}(u_S(q_t)) = \begin{cases} q_t \cdot 1 + (1 - q_t) \cdot x, & \text{if } q_t \geq \frac{1}{2} \\ q_t \cdot y, & \text{if } q_t < \frac{1}{2} \end{cases} \tag{3.1}$$

where $\mathbb{E}_{q_t}(u_S(q_t)) := \mathbb{E}_{q_t}(u_S(\omega_t, a(q_t)))$. In Figure 2, the Sender's interim expected payoff is plotted by the solid black line.

It is well-known from the literature on repeated games and Bayesian persuasion (Aumann & Maschler, 1995; Kamenica & Gentzkow, 2011) that within a period, the Sender's problem amounts to choosing a distribution over posteriors induced by the information structure to maximise her interim expected payoff. Let $\mu \in \Delta(\Delta(\Omega))$ denote such a distribution. Due to Bayesian updating, the feasible set of distributions over posteriors from which the Sender can choose comprises those satisfying Bayes plausibility: $\mathbb{E}_\mu(q_t) = p_t$.[11] The Sender's optimal static value is the concave

[11] A key result in Aumann and Maschler (1995) is that for a given distribution over posteriors such that they equal the prior in expectation, the Sender can always design an experiment that correlates the signals with the state in such a way that it induces the given distribution over posteriors.

envelope[12] of her interim expected payoff (3.1), evaluated at $p_t$:

$$\hat{V}(p_t) = \min\{p_t(1+x), x + p_t(1-x)\},$$

which is plotted by the dashed blue line in Figure 2.

Thus, the optimal information structure for the Sender is to not disclose anything if $p_t \geq 1/2$; if $p_t < 1/2$, she induces a posterior of $1/2$ with probability $2p_t$ and 0 with complementary probability. Under this information structure, the Receiver never takes the low action in the good state: when he takes the low action, his posterior belief is 0, i.e. he is certain that the state is bad. The high action is taken with probability $\mu(p_t) = 1$ if $p_t \geq 1/2$ and with probability $\mu(1/2) = 2p_t$ if $p_t < 1/2$. Indeed, subject to Bayes plausibility, the Sender designs an information structure that maximally leads the Receiver to believe that the state is good, thereby maximising the probability that the high action is taken within a period. This is because myopically, she has an ordinal preference for action $h$ over action $\ell$. Consistent with the literature, I will refer to choosing the information structure described above every period as the *greedy policy*.

## 3.2 The Dynamic Problem

### 3.2.1 Beliefs

From the previous subsection, we see that by choosing a distribution over posterior beliefs, the Sender indirectly chooses the probability with which an action is taken. This, in turn, influences the distribution of the next period's prior, due to the state transitions. This implies that when choosing a policy, the Sender chooses a (stochastic) sequence of players' beliefs. Hence, to solve the dynamic problem, we first need to track the effect of inducing a posterior in a period on the following period's prior.

Recall that $p_t$ and $q_t$ denote the common prior and posterior belief of the state being $G$ in period $t$, respectively. After the Receiver chooses an action, the state evolves based on the transition

[12] The concave envelope, or concavification, of a function $f$ is the smallest concave function greater than or equal to $f$.

probabilities described in (2.1). Specifically, the players' prior in period $t+1$ is given by:

$$p_{t+1}(q_t, a_t) = \begin{cases} q_t p_G^h + (1-q_t)p_B^h, & \text{if } a_t = h \\ q_t p_G^\ell + (1-q_t)p_B^\ell, & \text{if } a_t = \ell \end{cases} \tag{3.2}$$

Thus, the players' prior in a period is a conditional distribution over states given the entire public history, i.e. signals and actions taken in all previous periods.

Using the Receiver's best-response function, the above mapping from posteriors to priors can be re-written as:

$$p_{t+1}(q_t) = \begin{cases} f^h(q_t) \coloneqq p_B^h + q_t(p_G^h - p_B^h), & \text{if } q_t > 1/2 \\ f^\ell(q_t) \coloneqq p_B^\ell + q_t(p_G^\ell - p_B^\ell), & \text{if } q_t < 1/2 \end{cases} \tag{3.3}$$

and, if $q_t = 1/2$, the Receiver chooses action $a_t$ under the Sender-favourable tie-breaking convention, and then $p_{t+1} = f^{a_t}(1/2)$.

### 3.2.2 Bellman Equation

We can now use the above to write the Bellman equation. Let $V : [0,1] \to [0,1]$ denote the Sender's maximum payoff from the dynamic optimisation problem. This function is characterised by the solution to the following functional equation:

$$\begin{gathered} V(p_t) = \sup_{\mu \in \Delta(\Delta(\Omega))} \mathbb{E}_\mu\{(1-\delta)\mathbb{E}_{q_t}(u_S(q_t)) + \delta V(p_{t+1}(q_t))\} \\ \text{subject to } \mathbb{E}_\mu(q_t) = p_t \end{gathered} \tag{3.4}$$

where $p_{t+1}(q_t)$ is as defined in equation (3.3). In Appendix A.2, I show (following Renault, Solan, & Vieille, 2017) that the value function $V(\cdot)$ is concave over its domain. Since the Sender's current payoff and $f^a(q)$ are affine in $q$ for a fixed action $a$, concavity of $V(\cdot)$ implies that any collection of posteriors inducing the same action can be merged into a single posterior, their probability-weighted average, without lowering the Sender's payoff.[13] Hence, it is without loss to focus on policies that

[13] That is, the Sender sends a single signal whenever the original information structure would send any of the signals leading to these posteriors. The Receiver still takes the same action at the merged posterior, since each action is optimal on an interval of beliefs. The Sender's expected current payoff is unchanged, and tomorrow's prior after the merged signal is the average of the original ones, so by concavity of $V$ her continuation payoff cannot fall (see Corollary 3 in Appendix A.2).

have at most one posterior inducing $h$ and at most one inducing $\ell$.

Let $q^h$ and $q^\ell$ denote the posteriors that induce actions $h$ and $\ell$, respectively. Bayes plausibility implies that these need to be chosen such that $p = \alpha q^h + (1-\alpha)q^\ell$, where $\alpha := \mu(q^h) \in [0,1]$ is the probability with which the posterior $q^h$ is induced. Then, the Bellman equation (3.4) can be re-written as (suppressing the time index for the current prior):[14]

$$\begin{aligned} V(p) = \max_{\alpha, q^h} \{&(1-\delta)[\alpha(q^h + (1-q^h)x) + (1-\alpha)q^\ell y] + \delta\alpha V(f^h(q^h)) + \delta(1-\alpha)V(f^\ell(q^\ell))\} \\ &\text{subject to } p = \alpha q^h + (1-\alpha)q^\ell, \alpha \in [0,1], q^h \in [1/2, 1], q^\ell \in [0, 1/2] \end{aligned} \tag{3.5}$$

### 3.2.3 Properties of Optimal Policies

In this subsection, I characterise the properties of optimal policies, which greatly simplifies the problem.

The Bellman equation (3.5) shows that all histories leading to the same prior $p$ give the Sender the same continuation payoff regardless of the time period in which this prior is reached. Since the maximum in equation (3.5) is attained at every prior,[15] it is without loss to restrict attention to stationary policies, which choose $(\alpha, q^h)$ as a function of the current prior only.

In this regard, consider any stationary policy of the Sender, which specifies the $(\alpha(p), q^h(p))$ pair that the Sender should choose at each prior $p$ ($q^\ell(p)$ is determined by Bayes plausibility). Let $\phi(\cdot)$ denote the expected payoff to the Sender from following such a policy. The following lemma, which generalises Lemma 8 of Renault, Solan, and Vieille (2017), says that $(\alpha(p), q^h(p))$ is optimal if and only if $\phi(\cdot)$ is concave over $[0,1]$ and there is no profitable one-shot deviation to disclosing no information:

**Lemma 1** (Properties of optimal policies)**.** A stationary policy $(\alpha(p), q^h(p))$ of the Sender is optimal if and only if $\phi(\cdot)$ is concave over $[0,1]$ and the following is satisfied:

$$\phi(p) \geq \begin{cases} (1-\delta)(p + (1-p)x) + \delta\phi(f^h(p)), & \text{if } p \geq 1/2 \\ (1-\delta)py + \delta\phi(f^\ell(p)), & \text{if } p \leq 1/2 \end{cases} \tag{3.6}$$

[14] Given $(\alpha, q^h)$, $q^\ell$ is pinned down by Bayes plausibility and the constraint $q^\ell \in [0, 1/2]$ reflects the feasible $(\alpha, q^h)$ pairs. When $\alpha = 1$, $q^\ell$ is induced with probability 0 and its value is irrelevant.

[15] This follows from the continuity of $V$ (Claim 1) and the tie-breaking convention of footnote 9, which together make the Sender's interim payoff upper semi-continuous; see the discussion after equation (A.5) in Appendix A.2.

To summarise, we have shown that it is without loss to focus on stationary policies that induce at most one posterior for every action. Further, we have the above result, which is very useful: it can be used to verify the optimality of any candidate stationary policy.[16]

### 3.3 Two Dynamic Benchmarks

To facilitate a comparison to the existing literature, I analyse two special cases. Ely (2017), Renault, Solan, and Vieille (2017), and Ashkenazi-Golan et al. (2023) have all shown that in the binary-state, binary-action case, (i) if states evolve exogenously, and (ii) the Sender has cardinal state-independent preferences, then the greedy policy is optimal. I look at each of these in turn.

**Proposition 1** (Exogenous transitions)**.** If states evolve exogenously, then the greedy policy is optimal for all values of $(x, y) \in [0, 1]^2$.

In other words, if transitions are exogenous, then the greedy policy is optimal under weak ordinal preferences of the Sender. The reason for the above is as follows. Since actions do not affect future states, the expected prior tomorrow is the same whatever the Sender discloses today. Therefore, today's communication matters for the future only through the dispersion it creates in tomorrow's prior, which, since the continuation value is concave, can only lower it. The greedy policy induces the high action as often as possible, and in the binary case this static gain is always at least as large as any loss in continuation value.

**Proposition 2** (Cardinal state-independent preferences)**.** If $(x, y) = (1, 0)$, i.e. the Sender has cardinal state-independent preferences, then for all values of the transition probabilities, the greedy policy is optimal.

Intuitively, when the Sender only cares about the high action, a higher belief is just a means of getting that action. Holding the belief back for the future yields less than one-for-one, because it is discounted and diluted by state transitions. Thus, spending the belief now always wins.

[16] Lemma 1 generalises beyond the binary state, binary action case; see Lemma 20 in Appendix G.

Thus, Propositions 1 and 2 together generalise previous results: in the binary-state, binary-action case, if (i) states evolve exogenously, or (ii) the Sender has cardinal state-independent preferences, then the greedy policy is optimal.

# 4 Positive Externalities

In the dynamic game, the Sender has twin objectives: *within a period*, she wishes to maximise the probability of the high action, and *across periods*, she wants to maximise the probability of the good state. When there are positive externalities, her static and dynamic incentives align:

**Theorem 1** (Positive externalities)**.** Under positive externalities, the greedy policy is optimal for all values of $(x, y) \in [0, 1]^2$.

The proof, in Appendices B and E, uses Lemma 1 and proceeds in two parts. First, I show that the payoff induced by the greedy policy is concave in $p$ if and only if $x \geq \bar{x} \coloneqq \frac{\delta(p_B^\ell - p_B^h)}{1+\delta p_B^\ell - \delta p_G^h}$. Since $p_B^h > p_B^\ell$, $\bar{x} \leq 0$, and so the payoff from the greedy policy is concave for all $(x, y) \in [0, 1]^2$.

Second, I verify that the Sender does not gain from a one-shot deviation to no disclosure. The following observation provides a useful simplification. Since the Sender does not disclose any information when $p \geq 1/2$, this condition only needs to be verified when $p \in [0, 1/2)$. If the Sender one-shot deviates to no disclosure over this interval, the Receiver takes the low action, giving the Sender an expected payoff of $py$ from the current stage. This means that when $y = 1$, the Sender is maximally incentivised to deviate to no disclosure. Indeed, given $x$, the higher the value of $y$, the more the Sender prefers the good state. Thus, if the greedy policy is optimal when the Sender is most incentivised to deviate, then it is optimal for lower values of $y$ as well. Moreover, the Sender's long-run payoff under the greedy policy, and hence the one-shot deviation condition, is affine in $x$. Extending to all real $x$, it therefore suffices to verify the condition at the extreme points $x = \bar{x}$ and $x = 1$ as $[0, 1] \subseteq [\bar{x}, 1]$. When $x = \bar{x}$, this condition is satisfied for all $y \leq \bar{y} \coloneqq \frac{1+\delta p_B^\ell - \delta p_G^\ell}{1+\delta p_B^\ell - \delta p_G^h}$; since $p_G^h > p_G^\ell$, $\bar{y} \geq 1$. The rest of the proof proceeds to show that when $(x, y) = (1, 1)$, under positive externalities, the Sender does not gain from a one-shot deviation to no disclosure.

Thus, the Sender's weak ordinal preference for the high action and the good state is sufficient

for the greedy policy to be optimal under positive externalities.

# 5 Negative Externalities

In general, the Sender faces the following trade-off: the short-term gain from disclosing information about the current state, versus the benefit from influencing the evolution of the state, via the Receiver's action. Under negative externalities, her incentives today and tomorrow pull in opposite directions: the high action increases her static payoff, but it worsens the distribution of future states, and hence, lowers payoffs.

Recall that the Receiver wants to match his action to the state: he takes the high action when he believes the good state is more likely, and the low action otherwise. I refer to a mismatch between action and state, that is, the high action in the bad state or the low action in the good state, as a *mistake*. The only way the Receiver can avoid mistakes altogether is when the Sender reveals the state every period, or *full disclosure*.

As in static persuasion, the Sender gains from withholding information only if her interests *conflict* with the Receiver's, that is, if in some state she prefers the action he would not choose there. She does not value mistakes as such, but a conflict in a state is exactly what makes a mistake in that state worth inducing. In the stage game, the only conflict is in the bad state, where she weakly prefers the high action; in the good state, both prefer it. However, with intertemporal externalities, today's action also changes tomorrow's state, and this changes the Sender's effective preferences. In the bad state, the high action gives her $x$ today instead of 0, but makes the good state less likely tomorrow. In the good state, the low action costs her $1 - y$ today, but makes the good state more likely tomorrow.

To weigh these effects against each other, recall the thresholds from Section 4:

$$\bar{x} = \frac{\delta(p_B^\ell - p_B^h)}{1 + \delta p_B^\ell - \delta p_G^h}, \qquad \bar{y} = \frac{1 + \delta p_B^\ell - \delta p_G^\ell}{1 + \delta p_B^\ell - \delta p_G^h},$$

and suppose that from tomorrow on the Sender fully discloses the state. Then, compared with matching the action to the state, inducing the high action in the bad state today changes her

payoff by $(1-\delta)(x-\bar{x})$, and inducing the low action in the good state changes it by $(1-\delta)(y-\bar{y})$.[17] Since $1-\bar{y} = \delta(p_G^\ell - p_G^h)/(1+\delta p_B^\ell - \delta p_G^h)$, both $\bar{x}$ and $1-\bar{y}$ take the same form: the discounted change in tomorrow's probability of the good state, multiplied by $1/(1+\delta p_B^\ell - \delta p_G^h)$. This factor is the value of the good state: under full disclosure, starting a period in the good rather than the bad state is worth as much to the Sender as a one-off payoff of $1/(1+\delta p_B^\ell - \delta p_G^h)$ today.[18] So $\bar{x}$ is the future cost of the high action in the bad state, and $1-\bar{y}$ is the future benefit of the low action in the good state. The Sender's interests therefore conflict with the Receiver's in the bad state if and only if $x > \bar{x}$, and in the good state if and only if $y > \bar{y}$.

This also explains Theorem 1. Under positive externalities, $\bar{x} \leq 0$ and $\bar{y} \geq 1$: the externality reinforces the static conflict in the bad state and never creates one in the good state. The Sender behaves like a static Sender biased towards the high action, and the greedy policy is optimal. Under negative externalities, $\bar{x} \geq 0$ and $\bar{y} \leq 1$, and the externality works the other way: it can remove the conflict in the bad state and create one in the good state. In this sense, the externality determines whose side the Sender is on in each state.[19]

This calculation assumes full disclosure from tomorrow on. It therefore pins down when full disclosure is optimal, but not the whole solution: when interests conflict in both states, the Sender must also take into account where today's disclosure leaves tomorrow's belief. I return to this second force in Section 5.3.3.

## 5.1 Policies

Before characterising the optimal policy, I describe the policies that can be optimal. In Appendix A.2, I show that it is without loss to restrict attention to policies in which the set of priors at which the Sender communicates information is either empty, the interval $(0,1)$, or an interval with $1/2$ as an endpoint (see Lemmas 5 and 6). Thus, each such policy, indexed by $\pi$, can be described by a

[17] Under full disclosure, the Sender's payoff from prior $p$ is $\theta(p) = \frac{(1-\delta)p+\delta p_B^\ell}{1+\delta p_B^\ell - \delta p_G^h}$ (see Appendix D). In the bad state, the high action instead of the low action changes the payoff by $(1-\delta)x - \delta[\theta(p_B^\ell) - \theta(p_B^h)] = (1-\delta)(x-\bar{x})$; in the good state, the low action instead of the high action changes it by $(1-\delta)(y-1) + \delta[\theta(p_G^\ell) - \theta(p_G^h)] = (1-\delta)(y-\bar{y})$.

[18] This is because $\theta(1) - \theta(0) = \frac{1-\delta}{1+\delta p_B^\ell - \delta p_G^h}$.

[19] In Appendix F, I apply the same reasoning to mixed externalities, where the externalities have different signs in the two states.

*communication interval* $I_\pi = (q_\pi^\ell, q_\pi^h)$, with $q_\pi^\ell \leq 1/2 \leq q_\pi^h$:

- At any prior $p \in I_\pi$, the Sender splits the belief between the endpoints of the interval. She induces the posterior $q_\pi^h$, at which the Receiver takes action $h$, with probability $\alpha_\pi(p) = (p - q_\pi^\ell)/(q_\pi^h - q_\pi^\ell)$, and the posterior $q_\pi^\ell$, at which he takes action $\ell$, otherwise.
- At any prior $p \notin I_\pi$, the Sender discloses nothing. The Receiver takes action $h$ if $p > 1/2$, action $\ell$ if $p < 1/2$, and breaks ties in the Sender's favour at $p = 1/2$.

Policies differ in where the communication interval lies, and as a result, in which mistakes they induce:

| Policy | $I_\pi$ | $q_\pi^\ell$ | $q_\pi^h$ | Mistakes induced |
|---|---|---|---|---|
| Full disclosure | $(0, 1)$ | 0 | 1 | none |
| Greedy | $(0, 1/2)$ | 0 | $1/2$ | high action in the bad state |
| Judicious | $(1/2, 1)$ | $1/2$ | 1 | low action in the good state |
| Truncated greedy | $(\underline{q}, 1/2)$ | $\underline{q}$ | $1/2$ | both |
| Truncated judicious | $(1/2, \overline{q})$ | $1/2$ | $\overline{q}$ | both |

Table 1: Policies as a communication interval with two posteriors

*Full disclosure* fully reveals the state every period. While the *greedy policy* maximises the probability of the high action in every period, the *judicious policy* is its mirror image. In Figure 3, the black line is the Sender's interim payoff: her expected payoff from inducing the posterior $q$ today and following the judicious policy from tomorrow on. The blue line is its concave envelope, her long-run payoff from the judicious policy as a function of the prior $p$. Under the judicious policy, the Sender discloses nothing when $p_t \leq 1/2$, and when $p_t > 1/2$ she splits beliefs between $1/2$ and 1. The high action is never taken in the bad state: the high action follows a posterior belief of 1, when the state is good. This maximises the probability of the low action, which is taken with probability $\mu(p_t) = 1$ if $p_t \leq 1/2$ and with probability $\mu(1/2) = 2(1 - p_t)$ if $p_t > 1/2$. Equivalently, the judicious policy minimises the probability of the high action. In this sense, the judicious policy completely reverses the greedy policy.

The policies discussed so far—greedy, judicious and full disclosure—share a common feature: if the Sender discloses information, the Receiver is certain of at least one state with positive prob-

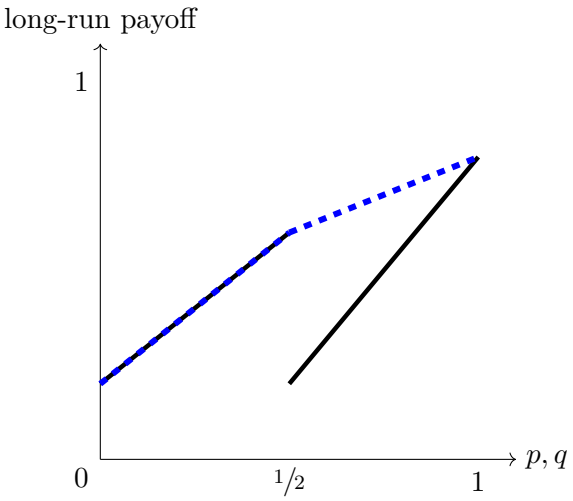



Figure 3: Sender's interim (black) and long-run (blue) expected payoff under the judicious policy

ability. The two *truncated* policies never make the Receiver certain of the state (Figure 4): each uses two interior posteriors. The truncated greedy policy truncates the communication interval from $(0, 1/2)$ to $(\underline{q}, 1/2)$ for some $\underline{q} \in (0, 1/2)$. At any prior between $\underline{q}$ and $1/2$, the Sender still raises the Receiver's belief to $1/2$ with some probability to obtain the high action today. However, she lowers the belief only to $\underline{q}$ instead of revealing the bad state, and below $\underline{q}$ she discloses nothing. The truncated judicious policy, with threshold $\overline{q} \in (1/2, 1)$, is defined symmetrically. Finally, if the communication interval is empty, i.e. $I_\pi = \emptyset$, the Sender never discloses information. This is the no-disclosure policy.

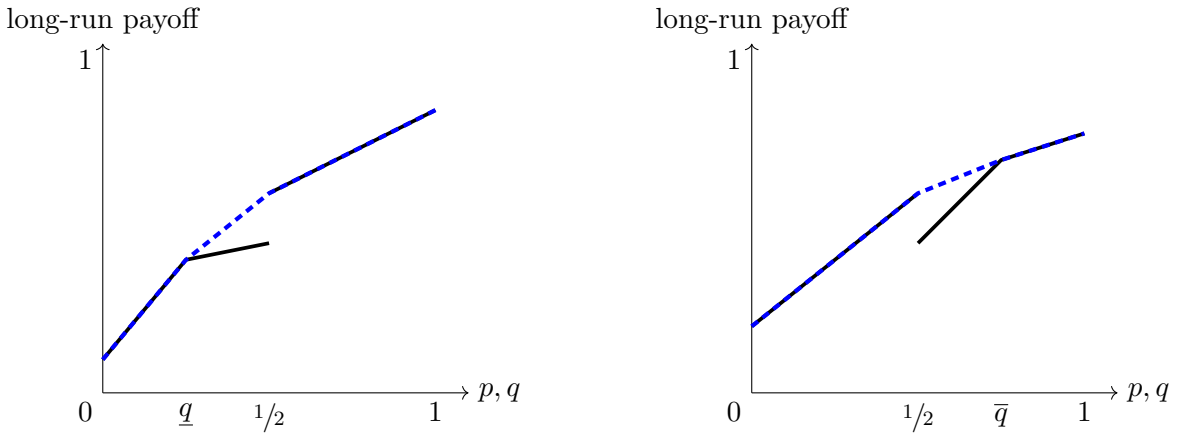
### 5.2 Characterisation

Recall that the Sender's interests conflict with the Receiver's in the bad state if and only if $x > \bar{x}$, and in the good state if and only if $y > \bar{y}$. The optimal policy is organised around these two comparisons:

**Theorem 2** (Negative externalities)**.** Under negative externalities:

(i) Full disclosure is optimal if and only if $(x, y) \leq (\bar{x}, \bar{y})$,

(ii) The greedy policy is optimal if and only if $x \geq \bar{x}$ and $y \leq \min\{1, L_g(x)\}$,




(a) Truncated greedy policy

(b) Truncated judicious policy

Figure 4: Sender's interim (black) and long-run (blue) expected payoff under the truncated greedy and truncated judicious policies

(iii) The judicious policy is optimal if and only if $y \geq \bar{y}$ and $x \leq \min\{1, L_j(y)\}$,

(iv) Otherwise, there exists a no-disclosure line, with the optimal policy being truncated greedy below it and truncated judicious above it.

The functions $L_g(\cdot)$, $L_j(\cdot)$ and the no-disclosure line are affine, strictly increasing, and pass through $(\bar{x}, \bar{y})$. Their exact functional form depends on the values of the parameters: transition probabilities and discount factor.

Figure 5 illustrates the theorem.[20] Its organising point is the corner $(\bar{x}, \bar{y})$: the Sender's interests conflict with the Receiver's in the bad state to its right, and in the good state above it. When interests are aligned in both states, she fully discloses the state. When they conflict only in the bad state, she induces the high action there as often as possible, which is the greedy policy; when they conflict only in the good state, she induces the low action there as often as possible, which is the judicious policy. These are the one-sided persuasion problems familiar from the static literature. When interests conflict in both states, the two conflicts compete. If one is much stronger, the Sender ignores the other, and the greedy or judicious policy remains optimal; $L_g$ and $L_j$ mark where the weaker conflict becomes strong enough to matter, and they slope upwards because a

[20] Figure 5 shows one of four possible configurations: the no-disclosure line may coincide with $L_g$, with $L_j$, with neither (as shown), or with both, in which case the truncated policies are never optimal.

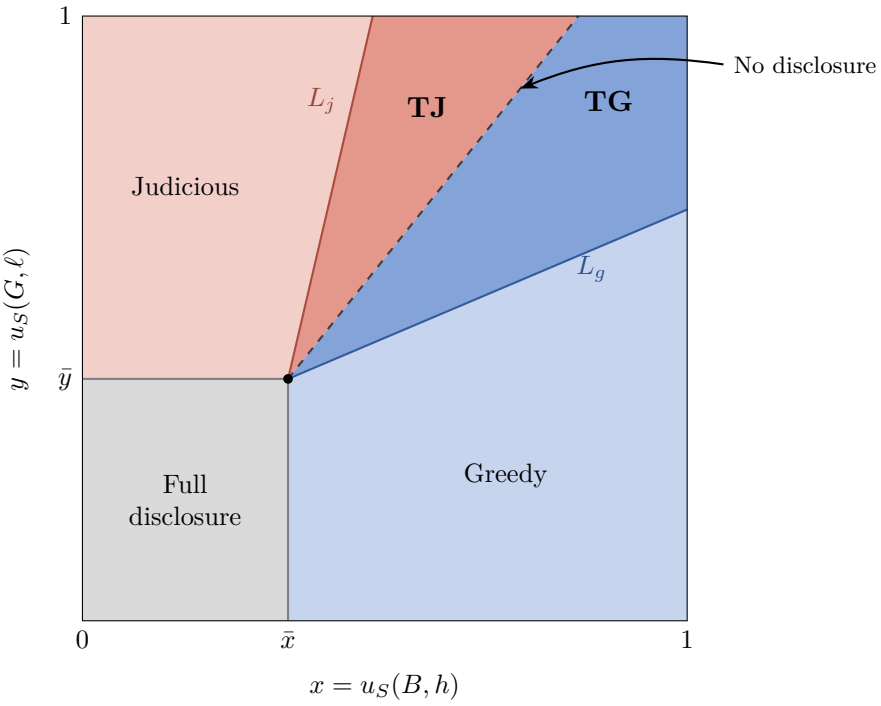



Figure 5: Optimal policies under negative externalities;
TG: truncated greedy; TJ: truncated judicious

stronger conflict in one state takes a stronger conflict in the other to outweigh it (Section 5.3.2). Only when both are strong does she induce both mistakes, through a truncated policy, so that the Receiver is never certain of the state. This interior pooling is the main qualitative difference from the existing literature: with exogenous transitions, the greedy policy is always optimal. Thus, under negative externalities, various policies are optimal depending on the cardinal features of the Sender's preferences, captured through the value of $(x, y)$.

## 5.3 Intuition and Proof Outline

This subsection explains the intuition behind each part of Theorem 2.

### 5.3.1 Aligned Interests: Full Disclosure

**Example 1.** Suppose $(x, y) = (0, 0)$, so that the Sender obtains a positive payoff only when the state is good and the Receiver takes the high action. Inducing the high action in the bad state

gives her nothing today and makes the good state less likely tomorrow, so she has no reason to do it. Inducing the low action in the good state is a genuine trade-off: it sacrifices her payoff of 1 today, but raises the probability that tomorrow's state is good from $p_G^h$ to $p_G^\ell$. By the computation above, the future benefit is worth $1 - \bar{y}$, which is less than the 1 sacrificed today, since $\bar{y} > 0$, that is, $\delta(p_G^\ell - p_B^\ell) < 1$. So, with full disclosure thereafter, the Sender's interests are aligned with the Receiver's in both states. Because the full disclosure payoff is affine in the prior, no other information structure in the current period does better either, and by Lemma 1, full disclosure is optimal. In particular, the judicious policy sacrifices more today than it gains in the future.

The same argument applies whenever $x \leq \bar{x}$ and $y \leq \bar{y}$ (see Appendix D). Under negative externalities and $\delta > 0$, both thresholds are positive, so this region always contains a neighbourhood of $(0, 0)$.

### 5.3.2 Conflict in One State: The Greedy and Judicious Policies

When $x > \bar{x}$ and $y \leq \bar{y}$, the Sender's interests conflict with the Receiver's only in the bad state: the high action there is worth more to her today than it costs her tomorrow. In the extreme case of cardinal state-independent preferences, $(x, y) = (1, 0)$, the Sender only cares about the high action, and the greedy policy is optimal for all values of the transition probabilities (Proposition 2). The mirror-image case is the following.

**Example 2.** Suppose $(x, y) = (0, 1)$, so that the Sender is indifferent between the two actions in the stage game, and any disclosure is driven purely by dynamic incentives. In both states she prefers the low action, because it makes the good state more likely tomorrow. In the bad state, this is also what the Receiver wants, so their interests are aligned; in the good state, it is not, so they conflict. She therefore induces the low action as often as possible, and the judicious policy is optimal.

The greedy and judicious policies remain optimal when interests also conflict in the other state, provided that this second conflict is weak. The two conflicts compete, because a policy that maximises the probability of the high action also minimises that of the low action. Under the

greedy policy, the Sender induces the high action whenever she can. As $y$ rises above $\bar{y}$, the conflict in the good state grows, and above $L_g(x)$ she prefers to give up some high actions in the bad state in order to induce some low actions in the good state. The higher $x$, the stronger the conflict in the bad state, and the higher $y$ must be before she makes this trade; hence $L_g$ is increasing. The same logic applies to the judicious policy and $L_j$.[21]

### 5.3.3 Conflict in Both States: The Truncated Policies

**Example 3.** Consider the following parameter specification: $\delta = 0.9$, $(x, y) = (1, 1)$, $p_G^h = 0.5$, $p_B^h = 0.1$, $p_G^\ell = 0.9$, and $p_B^\ell = 0.4$, so that $\bar{x} \approx 0.297$ and $\bar{y} \approx 0.604$. At the prior $p = 0.3$, the optimal information structure induces the two posteriors $q^h = 0.5$ and $q^\ell = 0.2$, with $\alpha = 1/3$. At $q^h = 0.5$, the Receiver is indifferent and, under the Sender-favourable tie-breaking convention, takes the high action; at $q^\ell = 0.2$, he takes the low action. The Sender's long-run payoff at $p = 0.3$ is:

| Policy | Long-run payoff at $p = 0.3$ |
|---|---|
| Truncated greedy policy | 0.6667 |
| Greedy policy | 0.6610 |
| Judicious policy | 0.5792 |
| Full disclosure | 0.4286 |

In the example above, $x > \bar{x}$ and $y > \bar{y}$, and the conflict in neither state is weak enough to ignore. At a prior between $\underline{q}$ and $1/2$, such as $p = 0.3$, the Sender splits the Receiver's belief between $1/2$ and $\underline{q} = 0.2$. To see why, recall that, with full disclosure from tomorrow onward, a mistake in the bad state is worth $(1 - \delta)(x - \bar{x})$ to the Sender and a mistake in the good state is worth $(1 - \delta)(y - \bar{y})$, relative to matching the action to the state. In the example, $x - \bar{x} \approx 0.70$ and $y - \bar{y} \approx 0.40$. At a posterior $q \geq 1/2$, the Receiver takes the high action, which results in a mistake with probability $1 - q$; at a posterior $q < 1/2$, he takes the low action, which results in a mistake with probability $q$.

[21] See Appendices B, C and E.4 for details.

**The upper posterior.** Under negative externalities, the high action makes the good state less likely tomorrow in both states, so it may seem that the Sender should induce it less often, by using an upper posterior above $1/2$. Suppose the Sender uses a higher upper posterior. This raises the probability of the low action, but the low action is taken at $\underline{q} < 1/2$, where the state is bad with probability $1 - \underline{q} > 1/2$. Thus, most of the increase falls in the bad state. In the bad state, the Sender prefers the high action, since its gain today, $x$, outweighs its cost tomorrow, $\bar{x}$. For each unit increase in the probability of the low action in the good state, the probability of the high action in the bad state falls by $(1 - \underline{q})/\underline{q}$ units, four in the example.[22] Thus, since $1/2$ is the lowest posterior at which the Receiver takes the high action, the Sender keeps the upper posterior at exactly $1/2$.

**The lower posterior.** With the upper posterior at $1/2$, the Receiver's chance of erring always equals his belief that the state is good: at $1/2$ he takes the high action, which is wrong half the time, and at the lower posterior he takes the low action, which is wrong with probability equal to the lower posterior. Since his beliefs average to the prior, he errs with probability $p$, whatever the lower posterior. Therefore, in the current period, the lower posterior determines only which mistake he makes. Under the greedy policy, where the lower posterior is 0, every mistake is the high action in the bad state. If the Sender discloses nothing, every mistake is the low action in the good state. Raising the lower posterior from 0 moves between these extremes, converting mistakes in the bad state into mistakes in the good state one-for-one. At $p = 0.3$, the greedy policy induces the high action in the bad state with probability 0.3. On the other hand, the truncated greedy policy induces the high action in the bad state with probability $1/6$, and the low action in the good state with probability $2/15$.[23]

**Tomorrow's belief.** If each mistake had a fixed value, the one-for-one conversion would lead the Sender to an extreme. This would be the greedy policy here, since $0.70 > 0.40$. However, these values assumed full disclosure from tomorrow on, under which the value of tomorrow's belief is

[22] For example, at $p = 0.3$, raising the upper posterior from $1/2$ to 0.6 raises the probability of the low action from $2/3$ to $3/4$. The probability of the low action in the good state therefore rises by $1/60$, and that of the high action in the bad state falls by $1/15$.

[23] By Bayes plausibility, the greedy policy induces the posterior $1/2$ with probability 0.6, and the state is then bad half the time. The truncated greedy policy induces $1/2$ with probability $1/3$ and 0.2 with probability $2/3$, giving $1/3 \cdot 1/2 = 1/6$ and $2/3 \cdot 0.2 = 2/15$. In both cases, the probabilities of the two mistakes sum to 0.3.

linear. Under the optimal policy it is not: its slope changes at $1/2$, where the Receiver's action switches. In the example, raising the lower posterior raises tomorrow's prior after the low action. While this prior is below $1/2$, raising it is valuable, since the closer it is to $1/2$, the more often the Sender can induce the high action tomorrow. Once it reaches $1/2$, the Receiver takes the high action without any disclosure, while as uncertain as possible. Raising it further gains less, since he takes the high action anyway, and in the example this no longer outweighs the cost today. The Sender therefore raises the lower posterior until tomorrow's prior is exactly $1/2$: $f^\ell(0.2) = 0.4 + 0.2(0.9 - 0.4) = 1/2$. This is a general feature of the truncated policies: starting from the prior $f^\ell(\underline{q})$, the belief either reaches exactly $1/2$, without disclosure along the way, or never returns to the communication interval.[24]

The truncated judicious policy makes the symmetric trade, starting from the judicious policy. Which of the two is optimal depends on which conflict is stronger: the truncated greedy policy below the no-disclosure line, and the truncated judicious policy above it. On the line the two are balanced, and disclosing no information at all is optimal.

**When are the truncated policies optimal?** The set of preferences at which none of the greedy, judicious and full disclosure policies is optimal is $\{(x, y) > (\bar{x}, \bar{y}) : y > L_g(x), x > L_j(y)\}$, and this set may be empty.[25] As Example 3 shows, the truncated greedy policy relies on the Sender's ability to change the next period's *default action*—the action the Receiver takes when he receives no new information—through the posterior at which she induces the low action today.[26] In the example, the low action at any posterior below 0.2 leaves tomorrow's prior below $1/2$, so the next Receiver takes the low action by default; at 0.2 or above, it leaves the prior at or above $1/2$, so he takes the high action. If tomorrow's default action is instead determined by today's action alone, the Sender cannot change it through the posterior.[27] Tomorrow's value is then affine in today's posterior for

[24] See Lemma 19 in Appendix E.4. Section 6 describes the resulting belief cycles.

[25] Lemmas 17 and 18 in Appendix E.4 give the necessary and sufficient conditions for non-emptiness.

[26] Recall from equation (3.2) that the period-$t+1$ prior $p_{t+1}$ is a function of the period-$t$ posterior $q_t$ and period-$t$ action $a_t$.

[27] This happens when: (a) the default action is always the high action, i.e. $f^h([1/2, 1])$ and $f^\ell([0, 1/2])$ both lie in $[1/2, 1]$; (b) the default action is always the low action, i.e. $f^h([1/2, 1])$ and $f^\ell([0, 1/2])$ both lie in $[0, 1/2]$; and (c) the default alternates, so that each action is followed by the other, i.e. $f^h([1/2, 1])$ lies in $[0, 1/2]$ and $f^\ell([0, 1/2])$ lies in $[1/2, 1]$. The remaining case, in which each action is its own default, is not possible under negative externalities as $f^\ell(1/2) - f^h(1/2) = (p^\ell_G - p^h_G + p^\ell_B - p^h_B)/2 > 0$.

each action, as under full disclosure, so each mistake has a fixed value to the Sender, and the one-for-one conversion between them leads her to one of the extremes.

### 5.3.4 Proof Outline

The proof of Theorem 2, covered in Appendices B–E, applies Lemma 1 to each candidate policy. The payoff from the greedy policy is concave if and only if $x \geq \bar{x}$, and since the one-shot deviation condition is affine in $(x, y)$, it binds along a line through $(\bar{x}, \bar{y})$, which is $L_g$; the judicious policy and $L_j$ are symmetric. In the remaining region, I show that the no-disclosure line exists, and characterise the thresholds $\underline{q}$ and $\overline{q}$.

## 5.4 Why is the Characterisation Asymmetric?

The judicious policy is the mirror image of the greedy policy, yet in Figure 5 the greedy region is larger, and $L_g$ and $L_j$ have different slopes. There are three reasons. First, the stage game favours the high action. Its static gain, $x$ in the bad state and $1 - y$ in the good state, can be as large as 1 in each state, whereas its dynamic costs, $\bar{x}$ and $1 - \bar{y}$, are discounted and sum to less than 1:

$$\bar{x} + (1 - \bar{y}) = \frac{\delta[(p_B^\ell - p_B^h) + (p_G^\ell - p_G^h)]}{1 + \delta p_B^\ell - \delta p_G^h} < 1$$

since this inequality is equivalent to $\delta(p_G^\ell - p_B^h) < 1$. Thus, $\bar{y} > \bar{x}$: the corner $(\bar{x}, \bar{y})$ lies above the 45-degree line, and the rectangle below and to its right, where the greedy policy is always optimal, is larger than the rectangle above and to its left, where the judicious policy is always optimal. Second, the externalities may differ in strength across states: $\bar{x}$ and $1 - \bar{y}$ are proportional to the externalities in the bad and the good state, respectively, with the same factor.

Third, even when the externalities are equally strong, the two actions may affect the state differently. The slopes of $L_g$ and $L_j$ depend not only on how much each action changes tomorrow's probability of the good state, but also on the levels of the transition probabilities, and the greedy policy relies on the high action while the judicious policy relies on the low action. So $L_g$ and $L_j$ need not be mirror images. The two actions affect the state symmetrically if the high action keeps the good state as often as the low action keeps the bad state, and turns the bad state good as

often as the low action turns the good state bad, that is, if $p_G^h + p_B^\ell = 1$ and $p_B^h + p_G^\ell = 1$. With such symmetric transitions, the externalities are also equally strong in the two states, so $\bar{x} = 1 - \bar{y}$, and $L_g$ and $L_j$ are mirror images of each other about the 45-degree line through $(\bar{x}, \bar{y})$. Thus, the second and third sources both disappear.[28]

The first source is the hardest to remove. With symmetric transitions, $\bar{y} - \bar{x}$ converges to $p_B^h / p_B^\ell$ as $\delta \to 1$, which is positive unless $p_B^h = 0$; so even a perfectly patient Sender typically faces a corner above the 45-degree line. The corner lies on it only in the limit in which, in addition, each mistake locks in the current state: the low action keeps the good state for sure and the high action keeps the bad state for sure ($p_G^\ell = 1$ and $p_B^h = 0$). In this limit, $(\bar{x}, \bar{y})$ converges to $(1/2, 1/2)$, the centre of the unit square, and none of the three sources of asymmetry remains.

### 5.5 Comparative Statics

How do the parameters affect the optimal policy? Recall the two conflicts: the Sender's interests conflict with the Receiver's in the bad state if and only if $x > \bar{x}$, and in the good state if and only if $y > \bar{y}$. The next result shows how the Sender's patience and the transition probabilities move these two thresholds.

**Proposition 3** (Comparative statics)**.** Under negative externalities, with $\delta \in (0, 1)$:

(i) *Patience:* $\bar{x}$ is strictly increasing and $\bar{y}$ is strictly decreasing in $\delta$.

(ii) *Transition probabilities after a mistake:* $\bar{x}$ is strictly decreasing in $p_B^h$ and does not depend on $p_G^\ell$; $\bar{y}$ is strictly decreasing in $p_G^\ell$ and does not depend on $p_B^h$.

(iii) *Transition probabilities after a matching action:* $\bar{x}$ and $\bar{y}$ are both strictly increasing in $p_G^h$ and in $p_B^\ell$.

By part (i), patience weakens the conflict in the bad state and strengthens the one in the good state. In other words, a more patient Sender needs a larger static gain before she prefers the high

[28] Relabelling the states, the actions and the belief ($G \leftrightarrow B$, $h \leftrightarrow \ell$, $p \leftrightarrow 1 - p$) leaves symmetric transitions unchanged and turns the greedy policy into the judicious policy and each mistake into the other. Since the slopes of $L_g$ and $L_j$ depend only on how often each mistake occurs under these policies and their one-shot deviations (Appendices B and C), $L_j$ is the reflection of $L_g$.

action in the bad state, and accepts a larger static loss to obtain the low action in the good state. So patience does not simply make the Sender more transparent: she reveals the bad state more readily, but is more willing to obscure the good one. In Figure 5, the corner $(\bar{x}, \bar{y})$ moves right and down, and the full disclosure region becomes wider and shorter.

Part (ii) concerns the transitions that follow a mistake, $p_B^h$ after the high action in the bad state and $p_G^\ell$ after the low action in the good state. Each affects only the threshold for its own mistake. A lower $p_B^h$ makes the high action more harmful in the bad state, and so weakens the conflict there. A higher $p_G^\ell$ makes the low action more beneficial in the good state, and so strengthens the conflict there.

Part (iii) concerns $p_G^h$ and $p_B^\ell$, the transition probabilities after an action that matches the state. Each has two effects. It changes the value of starting a period in the good rather than the bad state, $1/(1+\delta p_B^\ell - \delta p_G^h)$, which scales both $\bar{x}$ and $1-\bar{y}$, and it changes the externality in one state. A higher $p_G^h$ raises this value, so $\bar{x}$ rises; it also makes the low action less beneficial in the good state, which outweighs the higher value (since $1-\bar{y}<1$), so $\bar{y}$ rises too. Symmetrically, a higher $p_B^\ell$ lowers the value, so $\bar{y}$ rises, and makes the high action more harmful in the bad state, which outweighs the lower value (since $\bar{x}<1$), so $\bar{x}$ rises too. Either way, the corner moves up and to the right, and both conflicts weaken.

The transition probabilities also affect the slopes of $L_g$ and $L_j$, which are given in Appendices B and C.

## 6 Belief Dynamics and Disclosure Patterns

The previous two sections characterise which policy is optimal. This section shows that the optimal policy also determines *when* the Sender discloses. While the optimal policies are stationary, i.e. they depend on calendar time only through the current belief, the belief moves over time, partly because of the actions the Sender induces (equation (3.2)). Thus, except under full disclosure, the Sender may stay silent for some periods, during which Receivers act without new information. Each such silent stretch may end with a new disclosure, or last forever.

In this section, I assume that every transition probability lies in $(0,1)$: from either state, and

after either action, both states can occur in the following period. This implies that every belief reached after the first period lies in $(0,1)$.

### 6.1 From Policies to Belief Paths

Recall that each policy $\pi \in \{$full disclosure $(fd)$, greedy $(g)$, judicious $(j)$, truncated greedy $(tg)$, truncated judicious $(tj)\}$ in Section 5 can be described by a communication interval $I_\pi$. Also recall that, given the current action, the function mapping the current posterior to the next period's prior is given by $f^a$. Thus, outside $I_\pi$ when no information is disclosed, the following period's prior is given by $p_{t+1} = F_\pi(p_t) \coloneqq f^{a_\pi(p_t)}(p_t)$, where $a_\pi(p)$ is the action the Receiver takes at the prior $p$ when nothing is disclosed. Starting from prior $p_t$, if no information is disclosed for $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ periods,[29] the prior in period $t+n$ is given by $\underbrace{F_\pi(F_\pi \cdots (F_\pi}_{n \text{ times}}(p_t))) \coloneqq F_\pi^{(n)}(p_t)$, where $F_\pi^{(0)}(p) = p$.[30] In the communication interval $I_\pi$, the next prior is one of two *landing points*: $z_\pi^h \coloneqq f^h(q_\pi^h)$ or $z_\pi^\ell \coloneqq f^\ell(q_\pi^\ell)$.

For $a \in \mathcal{A}$, let the length of the *quiet spell* that follows a disclosure inducing $a$ be denoted by $n_\pi^a \coloneqq \min\{n \geq 0 : F_\pi^{(n)}(z_\pi^a) \in I_\pi\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$. I refer to a disclosure period together with the quiet spell that follows it as a *disclosure episode*. If the Sender induces $a$, a disclosure episode lasts $1 + n_\pi^a$ periods. Let $r_\pi^a \coloneqq F_\pi^{(n_\pi^a)}(z_\pi^a), n_\pi^a < \infty$ denote the prior at which disclosure resumes. Then:

**Lemma 2** (Belief support)**.** Fix a policy $\pi$ and an initial prior $p_1 \in (0,1)$ and let $\tau \coloneqq \min\{t : p_t \in I_\pi\}$.

(i) For $t \leq \tau$, $p_t = F_\pi^{(t-1)}(p_1)$. The Sender discloses nothing in $t < \tau$.

(ii) For every $t > \tau$, $p_t \in P_\pi \coloneqq P_\pi^h \cup P_\pi^\ell$, where $P_\pi^a \coloneqq \{F_\pi^{(n)}(z_\pi^a) : 0 \leq n \leq n_\pi^a\}$.

(iii) If $n_\pi^h, n_\pi^\ell < \infty$, then $P_\pi$ has at most $n_\pi^h + n_\pi^\ell + 2$ elements, and $(p_t)_{t>\tau}$ is a Markov chain on $P_\pi$. Its only random transitions occur at $r_\pi^h$ and $r_\pi^\ell$, which move to $z_\pi^h$ with probability $\alpha_\pi(r_\pi^a)$ and to $z_\pi^\ell$ otherwise.

[29] When $n = 0$, the interpretation is that information *is* disclosed in period $t$. When $n = \infty$, then information is *never* disclosed from period $t$ onward.

[30] To be clear, the drift in beliefs during quiet spells can move across regions where the high and low actions are optimal under no information.

Lemma 2 says that beliefs follow a simple path. Before the first disclosure, the belief drifts deterministically. Each disclosure induces one of the two posteriors, so the next prior is one of the two landing points, $z_\pi^h$ or $z_\pi^\ell$. From there, the belief again drifts without disclosure, for $n_\pi^a$ periods, until it re-enters the communication interval at $r_\pi^a$ and the Sender discloses again. So after the first disclosure, the belief only visits points on two paths, and the only randomness is which posterior each disclosure induces. The public history is therefore a sequence of disclosure episodes, whose lengths depend only on the actions induced.

### 6.2 When Does Disclosure Stop?

Each of the greedy, judicious and truncated policies uses $1/2$, where the Receiver is indifferent, as one of its two posteriors. I refer to the action he takes there as the *threshold action*. This is the high action under the greedy and truncated greedy policies, and the low action under the other two. Let $p_*^a$ denote the fixed point of $f^a$. Then:

**Proposition 4** (Termination)**.**

(i) *Greedy policy.* Suppose $p_G^h + p_B^h \geq 1$. Then from any prior, the Sender almost surely discloses in only finitely many periods. Afterwards the Receiver takes action $h$ in every period and the belief converges to $p_*^h \geq 1/2$.

(ii) *Judicious policy.* Suppose $p_G^\ell + p_B^\ell \leq 1$. Then from any prior, the Sender almost surely discloses in only finitely many periods. Afterwards the Receiver takes action $\ell$ in every period and the belief converges to $p_*^\ell \leq 1/2$.

The condition in (i) is $f^h(1/2) \geq 1/2$: one period of the high action taken at the threshold leaves the next, uninformed Receiver willing to take the high action again.[31] In this sense the high action is *self-sustaining*: once the Sender sends a message that only just persuades the Receiver, she never needs to speak again, and greedy disclosure is a transitional phenomenon. The condition in part (ii) says the same of the low action. On the other hand:

[31] Propositions 4 and 5 are about the greedy and judicious policies themselves: given a policy, beliefs evolve by the transition probabilities alone, so the results do not depend on $(x, y)$ or $\delta$. Under negative externalities, however, each policy is optimal for some preferences (Theorem 2), so every case in the two propositions describes the optimal dynamics for some $(x, y)$.

**Proposition 5** (Disclosure episodes)**.**

(i) *Greedy policy.* Suppose $p_G^h + p_B^h < 1$. Then from any prior, the Sender discloses in infinitely many periods. A disclosure that induces action $h$ is followed by another disclosure in the next period. A disclosure that induces action $\ell$ is followed by $n_g^\ell \in \mathbb{Z}_{\geq 0}$ periods of action $h$ without disclosure.

(ii) *Judicious policy.* Suppose $p_G^\ell + p_B^\ell > 1$. Then from any prior, the Sender discloses in infinitely many periods. A disclosure that induces action $\ell$ is followed by another disclosure in the next period. A disclosure that induces action $h$ is followed by $n_j^h \in \mathbb{Z}_{\geq 0}$ periods of action $\ell$ without disclosure.

The condition for disclosure cycles is that the threshold action undermines itself. Under the greedy policy, $p_G^h + p_B^h < 1$ means $f^h(1/2) < 1/2$: a Receiver who is persuaded just enough to take the high action depletes the state, so that the next Receiver, left uninformed, would take the low action. The Sender therefore cannot rest on a message that induces $h$ and must speak again in the following period. A message that induces the low action can work differently. The low action replenishes the state: if $p_B^\ell \geq 1/2$, the next Receivers start optimistic and take the high action without being told anything. The Sender stays silent while this optimism lasts, for $n_g^\ell$ periods, and speaks again once the drift under the high action has brought the belief back below $1/2$. (If $p_B^\ell < 1/2$, the Sender speaks again immediately after either message.) In this sense good news is spent at once, whereas bad news can be an investment that pays out in periods of the high action obtained for free. In the fisheries application, under greedy disclosure an assessment that leads fleets to fish at high effort is followed by another assessment the next season, whereas an assessment that leads them to restrain their effort is followed by possibly several seasons of high effort with no new assessment. The judicious policy has the mirror-image rhythm.

Propositions 4 and 5 imply the following:

**Corollary 1.** Under negative externalities, $p_G^\ell + p_B^\ell > p_G^h + p_B^h$. Hence at most one of the greedy and judicious policies has disclosure that eventually stops. Under full disclosure, disclosure never stops.

Under negative externalities, the low action makes the good state more likely than the high action does. So if the high action at $1/2$ leaves the next Receiver willing to take the high action, the low action at $1/2$ leaves him unwilling to take the low action, and vice versa. At most one of the two actions sustains itself.

The truncated policies have an additional feature. Under the truncated greedy policy, a disclosure that induces the posterior $\underline{q}$ leads to the prior $f^\ell(\underline{q})$. By Lemma 19 in Appendix E.4, the belief path that starts from this prior, absent further disclosure, either reaches exactly $1/2$ without first entering $I_{tg}$, or never returns to $I_{tg}$. In Example 3, it reaches $1/2$ at once: $f^\ell(0.2) = 1/2$. The truncated judicious policy is symmetric, starting from $f^h(\overline{q})$. Combined with Lemma 2, this gives the following:

**Corollary 2** (Truncated policies)**.** Under a truncated greedy or truncated judicious policy with $n^h_\pi, n^\ell_\pi < \infty$, $r^h_\pi = r^\ell_\pi \coloneqq r_\pi$. Every disclosure episode after the first therefore starts from the same prior $r_\pi$, and the actions they induce are i.i.d., with action $h$ induced with probability $\alpha_\pi(r_\pi)$.

Corollary 2 requires both quiet spells to be finite, so that the belief re-enters the communication interval after every disclosure. After a disclosure that induces the truncated posterior ($\underline{q}$ or $\overline{q}$), the belief then reaches exactly $1/2$. Since $1/2$ is an endpoint of the communication interval, the Sender does not disclose there, and the next Receiver takes the threshold action without new information. From then on, the belief follows the same path as after a disclosure that induces $1/2$, so both quiet spells end at $r_\pi$. Therefore, after the first disclosure, every disclosure episode starts from the same prior, whatever the previous one induced. Under the greedy and judicious policies, by contrast, the prior at which disclosure resumes generally depends on the last action induced, so actions are correlated across episodes.

**Example 3** (revisited)**.** Under the truncated greedy policy, $I_{tg} = (0.2, 1/2)$. A disclosure that induces the high action leads to the prior $f^h(1/2) = 0.3$, which lies in $I_{tg}$, so the Sender discloses again in the next period. A disclosure that induces the low action leads to the prior $f^\ell(0.2) = 1/2$. The next Receiver then takes the high action without new information, and the prior returns to $f^h(1/2) = 0.3$. So every disclosure episode starts from $r_{tg} = 0.3$ and induces the high action with

probability $\alpha_{tg}(0.3) = 1/3$. A disclosure that induces the high action is followed by another in the next period, and one that induces the low action is followed by one period of no disclosure.

When are both quiet spells finite? Under the truncated greedy policy, the answer depends only on $n^{\ell}_{tg}$. If $n^{\ell}_{tg}$ is finite, the belief passes through $1/2$ during the quiet spell after the low action, and then follows the same path as after a disclosure that induces $1/2$. So $n^{h}_{tg}$ is finite whenever $n^{\ell}_{tg}$ is. There are therefore two cases. If $n^{\ell}_{tg} < \infty$, disclosure recurs forever once it starts, as in Corollary 2. If $n^{\ell}_{tg} = \infty$, disclosure stops at the first disclosure that induces the low action. Every episode induces the low action with positive probability, so the Sender almost surely discloses in only finitely many periods.

The second case arises, for example, when the threshold action sustains itself, i.e. $p^{h}_{G} + p^{h}_{B} \geq 1$. After a high action at $1/2$, the belief then never falls below $1/2$ again, so neither quiet spell ends, and the Sender discloses at most once. The greedy policy, by contrast, can disclose several times before it stops, because the belief after a disclosure that induces the low action, $p^{\ell}_{B}$, can return to the communication interval. The truncated posterior instead sends the belief to $1/2$, where the high action sustains itself.

The truncated judicious policy is symmetric, with $n^{h}_{tj}$ in place of $n^{\ell}_{tg}$ and $p^{\ell}_{G} + p^{\ell}_{B} \leq 1$ in place of $p^{h}_{G} + p^{h}_{B} \geq 1$.

# 7 Conclusion

This paper studies an expert, the Sender, who informs a sequence of Receivers whose actions shape the state their successors inherit. These intertemporal externalities change what information is for: beyond steering today's action, the expert uses it to steer the state, and through it, tomorrow's actions. In effect, the externality determines, state by state, whether the expert's interests are aligned with the Receivers' or opposed to them.

When the externality is positive, the static and dynamic motives align, and the static benchmark survives intact: the greedy policy is optimal whatever the Sender's cardinal preferences. When it is negative, it can remove the conflict of interest in the bad state and create one in the good

state. The Sender then fully discloses the state if neither conflict is present, uses the greedy or the judicious policy if one dominates, and pools at interior beliefs if both are strong, never making the Receiver certain of the state. This interior pooling is the main departure from persuasion about an exogenous state, where the greedy policy is always optimal.

Three implications follow. First, an expert who cares about the future may withhold information because of that concern, not despite it: under negative externalities, she may hold back good news to restrain current activity. Second, patience does not simply make her more transparent: a more patient expert reveals bad states more readily but is more willing to obscure good ones. Third, the timing of disclosure is part of the optimal policy, since the content of one message determines when the next one comes. For a fisheries body, the frequency and conclusiveness of stock assessments are therefore outcomes of the optimal policy, and a natural object for empirical work.

The analysis suggests several directions for future research. One is to allow the Sender and the Receivers to hold different prior beliefs about the state. Another is to let the Sender inform several Receivers within a period, so that she must coordinate or miscoordinate their actions while also steering the state across periods. A third is to characterise optimal policies with finitely many states and actions; Appendix G takes a first step.

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# Appendix A Foundations

## A.1 Notation

I first introduce some notation used throughout the appendices.

Define $s_\omega \coloneqq p^h_\omega - p^\ell_\omega$, $\omega \in \Omega$. Thus, there are positive (negative) intertemporal externalities if $\forall \omega \in \Omega$, $s_\omega > 0$ ($s_\omega < 0$). State transitions are exogenous if $s_\omega = 0$ $\forall \omega \in \Omega$.

### A.1.1 Beliefs

Let $s^a \coloneqq p^a_G - p^a_B$, $a \in \mathcal{A}$ denote the slope of $f^a$. Since $f^a$ is affine and maps $[0,1]$ into itself, its set of fixed points is non-empty; let $p^a_*$ denote a fixed point of $f^a$, selected as follows.

$\underline{|\mathbf{s}^a| < \mathbf{1}.}$ If $|s^a| < 1$, then $f^a$ is a contraction with a unique fixed point, that is,

$$f^a(p^a_*) = p^a_* \iff p^a_* = \frac{p^a_B}{1 - s^a} \tag{A.1}$$

For each $a \in \mathcal{A}$, let $f^{a(n)}(p) \coloneqq \underbrace{f^a(f^a \cdots (f^a}_{n \text{ times}}(p)))$, $n \in \mathbb{N}$, and define $f^{a(0)}(p) = p$ and $f^{a(\infty)}(p) = \lim_{n\to\infty} f^{a(n)}(p)$. Starting from prior $p$, if no information is disclosed for $n$ periods and the Receiver repeatedly takes the same action $a$, the prior in period-$n+1$ is given by $f^{a(n)}(p)$. Using equation (A.1) and the fact that $f^{a(n)}(p) = f^a(f^{a(n-1)}(p))$, iterating the affine map $f^a$ gives:

$$f^{a(n)}(p) = p^a_*\{1 - (s^a)^n\} + (s^a)^n p \tag{A.2}$$

so that $f^{a(\infty)}(p) = p^a_*$.

$\underline{|\mathbf{s}^a| = \mathbf{1}.}$ There are two cases to consider:

(i) If $s^a = 1$, then $p^a_G = 1$ and $p^a_B = 0$. Thus, $f^a$ is the identity and does not have a unique fixed point. I set $p^a_* = 1/2$ in this case.

(ii) If $s^a = -1$, then $p^a_G = 0$ and $p^a_B = 1$. Thus, $f^{a(n)}(p)$ alternates between $p$ and $1-p$. Equation (A.1) still gives the unique fixed point $p^a_* = 1/2$.

In both cases, equation (A.2) still holds. Additionally, for all values of $|s^a| \in [0,1]$, we have that:

$$p^a_* \gtreqless 1/2 \iff f^a(1/2) \gtreqless 1/2 \tag{A.3}$$

### A.1.2 No Information Discounted Expected Payoff

Let $k^h(p) \coloneqq \mathbb{E}_p(u_S(\omega, h)) = p + (1-p)x = x + p(1-x)$ and $k^\ell(p) \coloneqq \mathbb{E}_p(u_S(\omega, \ell)) = py$ be the Sender's expected payoff from the high and low actions in the stage game, respectively, and let $\kappa^a$ denote the slope of $k^a(\cdot)$. Starting from prior $p$, when the Sender discloses no information for $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ periods and Receiver repeatedly takes the same action $a$ in those periods, the Sender's *no information discounted expected payoff* is denoted by $U^a(p,n)$. Define $U^a(p,0) = 0$. Then:

$$U^a(p,n) = \begin{cases} 0, & \text{if } n = 0 \\ \sum_{i=0}^{n-1} \delta^i k^a(f^{a(i)}(p)), & \text{if } n > 0 \end{cases}$$

Using (A.2), the above can be re-written as follows:

$$U^a(p,n) = \begin{cases} 0, & \text{if } n = 0 \\ k^a(p^a_*)\frac{1-\delta^n}{1-\delta} + (p - p^a_*)\kappa^a \frac{1-(\delta s^a)^n}{1-\delta s^a}, & \text{if } n \in \mathbb{N} \\ \frac{k^a(p^a_*)}{1-\delta} + \frac{(p-p^a_*)\kappa^a}{1-\delta s^a}, & \text{if } n = \infty \end{cases} \tag{A.4}$$

If no information is disclosed and the same action $a$ is taken repeatedly for $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ periods, let $m^a_n \coloneqq (1-\delta)\frac{\partial U^a(p,n)}{\partial p}$ denote the *marginal effect of a higher current prior on the Sender's*

*payoff over those $n$ periods*:

$$m_n^a = \begin{cases} 0, & \text{if } n = 0 \\ (1-\delta)\kappa^a \sum_{i=0}^{n-1} (\delta s^a)^i, & \text{if } n \in \mathbb{N} \\ \frac{(1-\delta)\kappa^a}{1-\delta s^a}, & \text{if } n = \infty \end{cases}$$

Finally, I adopt the convention that $\delta^\infty h \coloneqq \lim_{n\to\infty} \delta^n h = 0$ for any bounded $h$. Since $f^{a(n)}(p) \in [0,1]$ for every $n$, this sets $\delta^n f^{a(n)}(p) = 0$ whenever $n = \infty$.[32]

## A.2 Properties of the Value Function $V(\cdot)$

### A.2.1 Uniqueness

**Lemma 3.** $V$ is the unique bounded solution of equation (3.4).

*Proof.* For bounded $W$ on $[0,1]$, let $TW$ be the right-hand side of equation (3.4) with $W$ in place of $V$. Then $V = TV$, and $|W - W'| \leq c$ implies $|TW - TW'| \leq \delta c$. So $T$ is a contraction: $V$ is the unique bounded solution of Bellman (3.4), and $T^n W \to V$ uniformly for every bounded $W$. ■

### A.2.2 Concavity

**Lemma 4.** $V$ is concave over its domain.

*Proof.* The proof of the result is in Renault, Solan, and Vieille (2017), reproduced here. To show concavity, we must show that $V(p) \geq \lambda V(p') + (1-\lambda)V(p'')$, where $p = \lambda p' + (1-\lambda)p''$ for $\lambda \in [0,1]$. At belief $p$, the Sender can guarantee an expected payoff of at least $\lambda V(p') + (1-\lambda)V(p'')$ by inducing beliefs of $p'$ and $p''$ with probabilities $\lambda$ and $1-\lambda$ respectively, and then following the optimal policy at the resulting belief. The result then follows since by definition, $V(p)$ denotes the maximum payoff to the Sender at the prior $p$. ■

**Corollary 3.** It is without loss to restrict attention to distributions $\mu \in \Delta(\Delta(\Omega))$ whose support has at most 2 elements.

[32] When $s^a = -1$, the limit defining $f^{a(\infty)}(p)$ does not exist unless $p = 1/2$. This is immaterial below, because $f^{a(\infty)}$ enters only through products of the form $\delta^n f^{a(n)}(\cdot)$, which is set to zero at $n = \infty$.

*Proof.* Since $V(\cdot)$ is concave, if two or more posteriors induce the same action from the Receiver, the Sender weakly gains by inducing a single (combined) posterior that induces the said action. Since the Receiver's action space is binary, it is without loss for the Sender to induce at most two posteriors at any prior $p$. ■

### A.2.3 Continuity

For $q \in [0,1]$, let $w^a(q) \coloneqq (1-\delta)k^a(q) + \delta V(f^a(q))$ denote the Sender's payoff from inducing the posterior $q$ together with action $a \in \{h, \ell\}$ and then continuing optimally. Since $V$ is concave by Lemma 4 above, and $k^a$ and $f^a$ are affine, $w^h$ and $w^\ell$ are concave on $[0,1]$, and hence continuous on $(0,1)$. Define:

$$w(q) \coloneqq \begin{cases} w^\ell(q), & \text{if } q < 1/2 \\ w^h(q), & \text{if } q > 1/2 \\ \max\{w^\ell(1/2), w^h(1/2)\}, & \text{if } q = 1/2 \end{cases}$$

**Claim 1.** $V$ is 2-Lipschitz. Hence $V$ is continuous on $[0,1]$, and its one-sided derivatives on $(0,1)$ lie in $[-2,2]$.

*Proof.* By Lemma 3, it suffices to show: if $W$ is a function on $[0,1]$ that takes values in $[0,1]$ and is 2-Lipschitz, then $TW$ has the same two properties. Fix such a $W$. Let $w_W$ be $w$ with $W$ in place of $V$, so that $TW(p) = \sup_{\mathbb{E}_\mu(q)=p} \mathbb{E}_\mu[w_W(q)]$. Then $w_W$ and $TW$ take values in $[0,1]$. Since $y, |s^\ell| \leq 1$, for $q \geq 1/2$, $w_W(q) - w_W(0) \leq 1 \leq 2q$ and for $q < 1/2$, $w_W(q) - w_W(0) \leq (1-\delta)q + 2\delta q \leq 2q$. So $\mathbb{E}_\mu[w_W] \leq w_W(0) + 2p$ whenever $\mathbb{E}_\mu(q) = p$, and $TW(0) = w_W(0)$; hence $TW(p) - TW(0) \leq 2p$. Symmetrically, $TW(p) - TW(1) \leq 2(1-p)$. $TW$ is concave (as in Lemma 4), so for $p < p'$,

$$-2 \leq \frac{TW(1) - TW(p)}{1-p} \leq \frac{TW(p') - TW(p)}{p' - p} \leq \frac{TW(p') - TW(0)}{p'} \leq 2.$$ ■

### A.2.4 Communication Interval

By Bellman equation (3.5),

$$\begin{aligned} V(p) &= \max_{\alpha, q^h} \{\alpha w^h(q^h) + (1-\alpha) w^\ell(q^\ell)\} \\ \text{subject to } p &= \alpha q^h + (1-\alpha) q^\ell, q^h \in [1/2, 1], q^\ell \in [0, 1/2] \end{aligned} \tag{A.5}$$

so that $V \geq w$, and $V(p) = w(p)$ if and only if disclosing no information is optimal at $p$. The maximum in (A.5) is attained because $w$ is upper semi-continuous on $[0,1]$.[33]

**Lemma 5.** Let $I \coloneqq \{p \in [0,1] : V(p) > w(p)\}$ be the set of priors at which the Sender provides information. Either $I = \emptyset$, or $I = (\underline{p}, \overline{p})$ for some $\underline{p} \leq 1/2 \leq \overline{p}$ with $\underline{p} < \overline{p}$, and $V$ is affine on $[\underline{p}, \overline{p}]$. Moreover, if $\overline{p} = 1/2$ then $V(1/2) = w^h(1/2) > w^\ell(1/2)$, and if $\underline{p} = 1/2$ then $V(1/2) = w^\ell(1/2) > w^h(1/2)$.

*Proof.* At $p \in \{0,1\}$ no information can be disclosed, so $0, 1 \notin I$. On $(0,1)$, $V$ is continuous and $w$ is upper semi-continuous, so $I$ is open and is a disjoint union of open intervals. Let $(c,d)$ be one of them, and take $p \in (c,d)$. Let $(\alpha, q^h)$ attain the maximum in (A.5). By concavity of $V$,

$$V(p) = \alpha w^h(q^h) + (1-\alpha) w^\ell(q^\ell) \leq \alpha V(q^h) + (1-\alpha) V(q^\ell) \leq V(p),$$

so both inequalities hold with equality. Hence the posteriors used satisfy $V = w$, i.e. they lie outside $I$, so $q^\ell \leq c < p < d \leq q^h$, and $V$ is affine on $[q^\ell, q^h] \supseteq [c,d]$. Also $V(c) = w(c)$ and $V(d) = w(d)$, since $c, d \notin I$.

Suppose $d < 1/2$. Then $V(c) = w^\ell(c)$, $V(d) = w^\ell(d)$, and concavity of $w^\ell$ gives $w^\ell \geq V$ on $[c,d]$, contradicting $(c,d) \subseteq I$. The same contradiction arises if $d = 1/2$ and $V(1/2) = w^\ell(1/2)$. So if $d \leq 1/2$, then $d = 1/2$ and $V(1/2) = w^h(1/2) > w^\ell(1/2)$. Symmetrically, if $c \geq 1/2$, then $c = 1/2$ and $V(1/2) = w^\ell(1/2) > w^h(1/2)$. Every other interval satisfies $c < 1/2 < d$. Since the intervals are disjoint, and the two conditions on $V(1/2)$ are mutually exclusive, $I$ contains at most one of them. ■

Thus, an optimal policy is to disclose nothing outside $I$, and to split every prior in $I$ between $\underline{p}$ and $\overline{p}$; by (A.5), this policy attains $V$ at every prior. In particular, if $\underline{p} = 0$ and $\overline{p} = 1/2$ it is the greedy

[33] Each $w^a$ is continuous, since $V$ is continuous (Claim 1). As $w(1/2) = \max\{w^\ell(1/2), w^h(1/2)\}$, $w$ is upper semi-continuous.

policy, if $\underline{p} = 1/2$ and $\overline{p} = 1$ it is the judicious policy, and if $\underline{p} = 0$ and $\overline{p} = 1$ it is full disclosure. If $\underline{p} \in (0, 1/2)$ and $\overline{p} = 1/2$ it is the truncated greedy policy with $\underline{q} = \underline{p}$, and if $\underline{p} = 1/2$ and $\overline{p} \in (1/2, 1)$ it is the truncated judicious policy with $\overline{q} = \overline{p}$ (see Table 1).

**Lemma 6.** If $1/2 \in I$, then full disclosure is optimal.

*Proof.* Suppose $1/2 \in I$. By Lemma 5, $I = (\underline{p}, \overline{p})$ with $\underline{p} < 1/2 < \overline{p}$, $V$ is affine on $[\underline{p}, \overline{p}]$, and $V = w$ on $[0, 1] \setminus I$. Since $V$, $w^h$ and $w^\ell$ are concave (Lemma 4), they have non-increasing left and right derivatives on $(0, 1)$, with $V'_- \geq V'_+$. For a closed interval $[c, d] \subset (0, 1)$, possibly a single point, I refer to $V'_-(c) - V'_+(d) \geq 0$ as the *slope drop* of $V$ across $[c, d]$. I refer to $[c, d]$ as *short* if $d - c < \overline{p} - \underline{p}$.

I first show that, for every short $[c, d] \subset (0, 1)$, there exists a short $[c', d'] \subseteq f^h([0, 1]) \cup f^\ell([0, 1])$ such that

$$V'_-(c) - V'_+(d) \leq \delta \max_{a \in \mathcal{A}} |s^a| \big[V'_-(c') - V'_+(d')\big] \tag{A.6}$$

A short interval cannot meet both $[0, \underline{p}]$ and $[\overline{p}, 1]$. If it meets neither, it lies in $I$, where $V$ is affine, so its slope drop is zero and (A.6) holds trivially. Suppose it meets $[0, \underline{p}]$; the other case is symmetric, with $h$ in place of $\ell$. Since $V'_+$ is constant on $[\underline{p}, \overline{p})$, replacing $d$ by $\min\{d, \underline{p}\}$ does not change the slope drop, so we may assume $[c, d] \subseteq (0, \underline{p}]$. On $[0, \underline{p}]$, $V = w = w^\ell$, because $\underline{p} < 1/2$. So $V'_-(c) = (w^\ell)'_-(c)$, and $V'_+(d) \geq (w^\ell)'_+(d)$. The latter holds with equality if $d < \underline{p}$, and as an inequality if $d = \underline{p}$, because $V \geq w^\ell$ on $[\underline{p}, 1/2)$ with equality at $\underline{p}$. Hence

$$V'_-(c) - V'_+(d) \leq (w^\ell)'_-(c) - (w^\ell)'_+(d) = \delta |s^\ell| \big[V'_-(c') - V'_+(d')\big], \qquad [c', d'] \coloneqq f^\ell([c, d])$$

The equality holds because $w^\ell(p) = (1 - \delta) k^\ell(p) + \delta V(f^\ell(p))$ with $k^\ell$ and $f^\ell$ affine, so $(w^\ell)'_\pm(p) = (1 - \delta)\kappa^\ell + \delta s^\ell V'_\pm(f^\ell(p))$. (If $s^\ell < 0$, the two sides of $V'$ are swapped, because $f^\ell$ reverses direction.) Finally, $d' - c' = |s^\ell|(d - c)$, so $[c', d']$ is short.

Let $\xi$ be the supremum of the slope drop of $V$ over short intervals in $(0, 1)$; by Claim 1, $\xi \leq 4$.[34] If $s^a \neq 0$, $f^a$ maps $(0, 1)$ into $(0, 1)$, so equation (A.6) gives $\xi \leq \delta\xi$; if $s^a = 0$, $w^a$ is affine and the slope drop in equation (A.6) is zero. Hence, $\xi = 0$.

[34] For general $(x, y) \in \mathbb{R}^2$, the same argument, with stage payoffs in $[\underline{u}, \overline{u}]$ where $\underline{u} \coloneqq \min\{0, x, y\}$ and $\overline{u} \coloneqq \max\{1, x, y\}$, shows that $V$ is Lipschitz with constant $\max\{|1 - x|, |y|, 2(\overline{u} - \underline{u})\}$. Lemma 6 only uses that this constant is finite.

Now split any $[c,d] \subset (0,1)$ into finitely many consecutive short intervals. Since $V'_- \geq V'_+$ at each point where two consecutive intervals meet, the slope drop across $[c,d]$ is at most the sum of the slope drops across the pieces, which is zero. So $V'_-(c) = V'_+(d)$ for all $c \leq d$ in $(0,1)$, and $V$ is affine on $(0,1)$.

It remains to show that $I = (0,1)$. Suppose $\underline{p} > 0$. Since $V$ is affine on $(0,1)$ and continuous on $[0,1]$, it is affine on $[0,1]$, so $w^\ell$ is affine on $[0,1]$. It equals $V$ on $(0,\underline{p}]$, so it equals $V$ on $(0,1)$. This contradicts $V > w = w^\ell$ on $(\underline{p}, 1/2) \subseteq I$. Hence $\underline{p} = 0$, and symmetrically $\overline{p} = 1$. So $I = (0,1)$, and the optimal policy splits every prior between the posteriors 0 and 1: full disclosure is optimal. ■

The contrapositive of the previous lemma gives the following:

**Corollary 4.** If full disclosure is not optimal, then $1/2 \notin I$, i.e. $V(1/2) = \max\{w^h(1/2),\, w^\ell(1/2)\}$.

To summarise, we have shown that: (i) the value function $V(\cdot)$ is unique, concave and continuous, (ii) it is without loss to induce at most one posterior for each action, (iii) the set of priors at which the Sender communicates information is an interval, where at every prior she splits the belief between the interval's endpoints, and she discloses nothing outside it, and (iv) if $1/2$ lies in the interior of this interval, then full disclosure is optimal.

### A.3 Proof of Lemma 1

Consider a stationary policy $(\alpha(p), q^h(p))$ of the Sender and let $\phi(\cdot)$ denote the Sender's payoff from using this policy.

( $\implies$ ): If $(\alpha(p), q^h(p))$ is optimal then $\phi(p) = V(p)$ and by Lemma 4, $\phi(\cdot)$ must be concave. Additionally, $(\alpha(p), q^h(p))$ must give a higher payoff than all deviations, including the one-shot deviation to providing no information in the current period, and then returning to the prescribed policy from the next period onward. This is precisely the inequality in (3.6).

( $\impliedby$ ): By definition, the payoff from any policy is at most the maximum, so that:

$$\phi(p) \leq \max_{\tilde{\alpha}, \tilde{q}^h} \{(1-\delta)[\tilde{\alpha}(\tilde{q}^h + (1-\tilde{q}^h)x) + (1-\tilde{\alpha})\tilde{q}^\ell y] + \delta\tilde{\alpha}\phi(f^h(\tilde{q}^h)) + \delta(1-\tilde{\alpha})\phi(f^\ell(\tilde{q}^\ell))\}$$

Let $\phi(\cdot)$ be concave in $p$ and (3.6) hold. Consider any arbitrary Bayes plausible experiment

$(\tilde{\alpha}(p), \tilde{q}^h(p))$. Then:

$$\begin{aligned}
&\tilde{\alpha}\left[(1-\delta)(\tilde{q}^h + (1-\tilde{q}^h)x) + \delta\phi(f^h(\tilde{q}^h))\right] + (1-\tilde{\alpha})\left[(1-\delta)\tilde{q}^\ell y + \delta\phi(f^\ell(\tilde{q}^\ell))\right] \\
&\quad \leq \tilde{\alpha}\phi(\tilde{q}^h) + (1-\tilde{\alpha})\phi(\tilde{q}^\ell) && \text{(by (3.6))} \\
&\quad \leq \phi\Big(\tilde{\alpha}\tilde{q}^h + (1-\tilde{\alpha})\tilde{q}^\ell\Big) && \text{(concavity)} \\
&\quad = \phi(p) && \text{(Bayes plausibility)}
\end{aligned}$$

Thus, $\phi(p)$ solves the Bellman equation (3.4). Since equation (3.4) has a unique bounded solution by Lemma 3, $\phi = V$ and $(\alpha(p), q^h(p))$ is optimal. ■

# Appendix B The Greedy Policy

Recall that the greedy policy maximises the probability with which the high action is taken by the Receiver each period, i.e.:

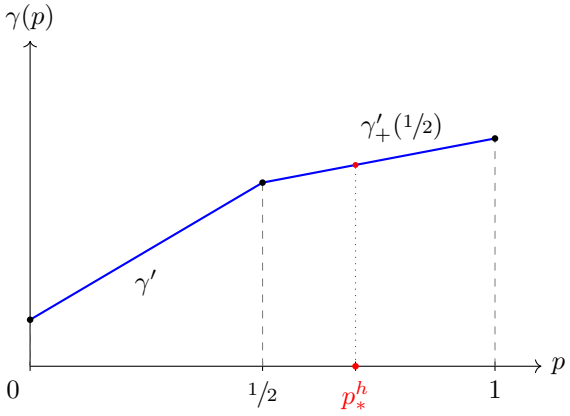
$$(\alpha_g(p), q_g^h(p)) = \begin{cases} (1, p), & p \geq 1/2 \\ (2p, 1/2), & p < 1/2 \end{cases}$$

Let $\gamma : [0,1] \to [0,1]$ denote the Sender's long-run payoff from using the greedy policy:

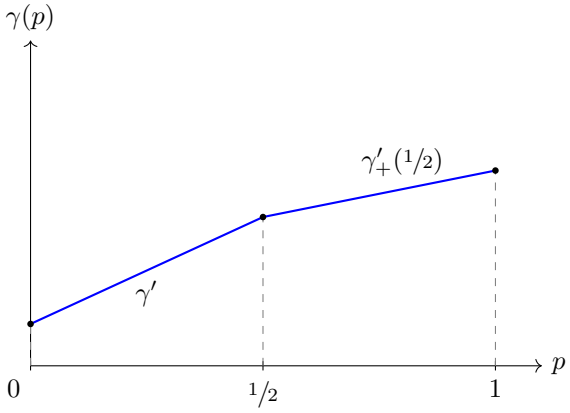
$$\gamma(p) = \begin{cases} (1-\delta)k^h(p) + \delta\gamma(f^h(p)), & p \geq 1/2 \\ 2p\gamma(1/2) + (1-2p)\gamma(0), & p < 1/2 \end{cases} \tag{B.1}$$

When $p = 0$, the Receiver is certain that the state is bad, so he chooses the low action. This implies that the next period's prior is $f^\ell(0) = p_B^\ell$, so that $\gamma(0) = \delta\gamma(p_B^\ell)$. Observe that, for $p < 1/2$, $\gamma$ is affine, with a slope of $\gamma' \coloneqq \gamma'_-(1/2) = 2[\gamma(1/2) - \gamma(0)]$.

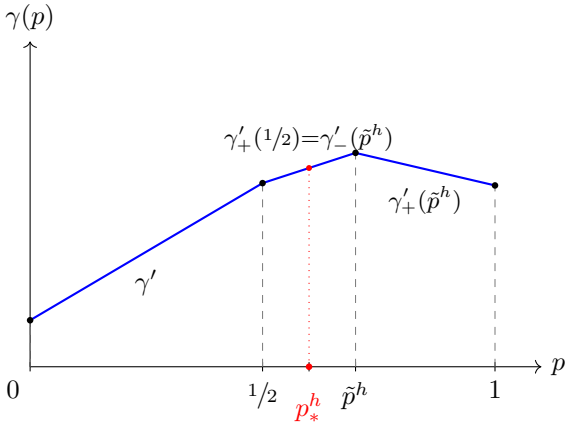
For given values of transition probabilities and preference parameters $x$ and $y$, to show that the greedy policy is optimal, we need to verify that both the conditions of Lemma 1 are satisfied. That is, we must show that $\gamma$ is concave in $p$ and that the Sender does not strictly gain from a one-shot deviation to no disclosure. I next analyse these in turn.



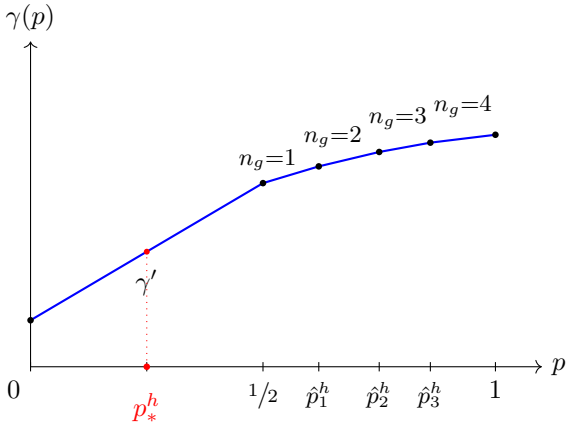

(a) Absorbing case: $f^h([1/2, 1]) \subseteq [1/2, 1]$




(b) Reverting case: $f^h([1/2, 1]) \subseteq [0, 1/2)$




(c) Partially absorbing case: $f^h(1/2) \geq 1/2 > f^h(1)$




(d) Cascading case: $f^h(1/2) < 1/2 \leq f^h(1)$

Figure 6: The Sender's payoff $\gamma(\cdot)$ under the greedy policy in each of the four cases of Section B.1; $p^h_*$ is the fixed point of $f^h(\cdot)$, drawn for $|s^h| < 1$

## B.1 Concavity

First, I analyse the payoff from the greedy policy. I start by showing the following:

**Lemma 7.** The payoff from the greedy policy is concave if and only if $\gamma' \geq m^h_\infty$.

*Proof.* The proof is divided into 4 cases (see Figure 6), depending on the range of values of $f^h([1/2, 1])$.[35] In each of these, I argue that $\gamma$ is piecewise affine. This means that for $\gamma$ to be concave, the left-hand derivative must exceed the right-hand derivative at the kinks, i.e. $\gamma'_-(p) \geq \gamma'_+(p)$ at kink $p$. It turns out that this is equivalent to showing that $\gamma' \geq m^h_\infty$.

**Case 1:** $\underline{f^h([1/2, 1]) \subseteq [1/2, 1].}$ The interval $[1/2, 1]$ is absorbing: because no information is disclosed, once the belief is greater than $1/2$, it stays at or above $1/2$ forever (see panel (a) of Figure 6). For $p \geq 1/2$, since the static payoff $k^h(\cdot)$ and the future belief $f^h(\cdot)$ are affine in $p$, $\gamma(\cdot)$ is affine on $[1/2, 1]$. Differentiating the first line of equation (B.1) with respect to $p$:

$$\gamma'_+(1/2) = (1-\delta)\kappa^h + \delta s^h \gamma'_+(1/2) \iff \gamma'_+(1/2) = m^h_\infty \tag{B.2}$$

It follows that $\gamma$ is concave if and only if $\gamma' \geq m^h_\infty = \gamma'_+(1/2)$.

**Case 2:** $\underline{f^h([1/2, 1]) \subseteq [0, 1/2).}$ When $p \geq 1/2$, the Sender discloses no information and $f^h(p) < 1/2$, so $\gamma'_+(1/2) = m^h_1 + \delta s^h \gamma'$. Thus, $\gamma$ is concave if and only if $\gamma' \geq \gamma'_+(1/2)$, or equivalently, $\gamma' \geq m^h_\infty$ (see panel (b) of Figure 6).

**Case 3:** $\underline{f^h(1/2) \geq 1/2 > f^h(1).}$ Since $f^h$ is affine, $f^h(1/2) > f^h(1)$ implies that $f^h$ is decreasing, i.e. $s^h < 0$. And since $f^h([1/2, 1])$ is an interval containing $1/2$, by the intermediate value theorem, there exists a unique $\tilde{p}^h \in [1/2, 1]$ such that $f^h(\tilde{p}^h) = 1/2$. This means that for $p \in [1/2, \tilde{p}^h]$, $f^h(p) \in [1/2, f^h(1/2)]$. Since $f^h(1/2) \leq \tilde{p}^h$, $[1/2, \tilde{p}^h]$ maps into itself. If $\tilde{p}^h > 1/2$, then $\gamma'_+(1/2)$ is given by equation (B.2) and $\gamma'_+(1/2) = m^h_\infty = \gamma'_-(\tilde{p}^h)$ (see panel (c) of Figure 6). If $\tilde{p}^h = 1/2$, then $\gamma'_+(1/2) = \gamma'_+(\tilde{p}^h)$, which is described next.

[35] Given the period-$t$ posterior $q_t$, the prior in period $t+1$ lies either in the interval $f^h([1/2, 1])$ or in the interval $f^\ell([0, 1/2])$. The linearity of $f^h$ and $f^\ell$, and the fact that $p^h_G$, $p^h_B$, $p^\ell_G$, $p^\ell_B$ are probabilities implies that $f^h([1/2, 1])$ and $f^\ell([0, 1/2])$ are interval subsets of $[0, 1]$.

By the definition of $\tilde{p}^h$, for $p > \tilde{p}^h$, $f^h(p) < 1/2$. Hence $\gamma'_+(\tilde{p}^h)$ is obtained by applying the first-period recursion to the affine branch below $1/2$, giving $\gamma'_+(\tilde{p}^h) = m_1^h + \delta s^h \gamma'$. It follows that $\gamma$ is concave if and only if $\gamma' \geq \gamma'_+(1/2) = \gamma'_-(\tilde{p}^h) \geq \gamma'_+(\tilde{p}^h)$. These inequalities reduce to $\gamma' \geq m_\infty^h$, using the fact that $s^h < 0$.

**Case 4:** $\underline{f^h(1/2) < 1/2 \leq f^h(1).}$ Since $f^h$ is affine, $f^h(1/2) < f^h(1)$ implies that $f^h$ is increasing, i.e. $s^h > 0$. As in case 3, by the intermediate value theorem, there exists a unique $\hat{p}_1^h \in [1/2, 1]$ such that $f^h(\hat{p}_1^h) = 1/2$. Thus, for $p \in [1/2, \hat{p}_1^h)$, $f^h(p) < 1/2$.

Suppose $p \geq \hat{p}_1^h$ so that $f^h(p) \geq 1/2$. Again, the Sender discloses no information, so the new belief is $f^h(f^h(p))$, which could be less than or greater than $1/2$. Define $\hat{p}_2^h \in [\hat{p}_1^h, 1]$ to be the point such that $f^h(\hat{p}_2^h) = \hat{p}_1^h$, so that for $p \geq \hat{p}_2^h$, $f^h(f^h(p)) \geq 1/2$. Continuing with the same logic, we can define a sequence of thresholds $1/2 =: \hat{p}_0^h < \hat{p}_1^h < \hat{p}_2^h < \cdots < \hat{p}_{\eta-1}^h < 1$, where $f^h(\hat{p}_n^h) = \hat{p}_{n-1}^h$, and set $\hat{p}_\eta^h \coloneqq 1$, where from an initial belief of $p = 1$, $\eta \in \mathbb{N}$ is the number of periods in which no information is disclosed before the belief falls below $1/2$. Since $f^h$ is a contraction,[36] this sequence is finite and since $f^h(1/2) < 1/2$, beliefs always slide towards $p_*^h$, which is less than $1/2$ by equation (A.3). For example, in panel (d) of Figure 6, $\eta = 4$.

It follows that $[1/2, 1] = [\hat{p}_0^h, \hat{p}_\eta^h]$ is the union of disjoint intervals: $[\hat{p}_0^h, \hat{p}_1^h) \cup [\hat{p}_1^h, \hat{p}_2^h) \cup \cdots \cup [\hat{p}_{\eta-1}^h, \hat{p}_\eta^h]$. For $p \in [\hat{p}_{n-1}^h, \hat{p}_n^h)$, $n = \min\{i : f^{h(i)}(p) < 1/2\}$, where $n \in \{1, 2, \ldots, \eta\}$. Hence, for $n \in \{0, 1, \ldots, \eta - 1\}$, $\gamma'_+(\hat{p}_n^h) = m_{n+1}^h + (\delta s^h)^{n+1} \gamma' = \gamma'_-(\hat{p}_{n+1}^h)$.

Thus, $\gamma(\cdot)$ is concave if and only if $\gamma' \geq \gamma'_+(\hat{p}_0^h) = \gamma'_-(\hat{p}_1^h) \geq \gamma'_+(\hat{p}_1^h) = \gamma'_-(\hat{p}_2^h) \geq \cdots \geq \gamma'_-(\hat{p}_\eta^h)$. These inequalities reduce to $\gamma' \geq m_\infty^h$, using the fact that $s^h > 0$. This completes the proof. ■

The previous lemma expresses the concavity of the greedy policy as a condition on its slope. This can be further reduced to get a simple condition on the preference parameter $x$. The following claims provide some useful steps:

**Claim 2.** Consider any prior $p \in [0, 1]$. Let $n_g(p) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ be the number of periods in which no information is disclosed before the belief falls below $1/2$, i.e. $n_g(p) = \min\{j : f^{h(j)}(p) < 1/2\}$. Using

[36] This is because $s^h = 1$ forces $p_G^h + p_B^h = 1$ and hence $f^h(1/2) = 1/2$, contradicting $f^h(1/2) < 1/2$. Thus, $s^h < 1$ and $f^h$ is a contraction.

equation (A.4), for $p \in (0,1]$, the long-run payoff from the greedy policy (B.1) can be re-written as:

$$\gamma(p) = (1-\delta)U^h(p, n_g(p)) + \delta^{n_g(p)}\left\{2f^{h(n_g(p))}(p)\gamma(1/2) + (1-2f^{h(n_g(p))}(p))\gamma(0)\right\} \tag{B.3}$$

where $\gamma(0) = \delta\gamma(p_B^\ell)$.

*Proof.* If $n_g(p) = 0$, equation (B.3) reduces to the same expression for $\gamma(\cdot)$ when $p < 1/2$ as defined in equation (B.1). If $n_g(p) \in \mathbb{N}$, then by assumption, $f^{h(n_g(p)-1)}(p) \geq 1/2 > f^{h(n_g(p))}(p)$. Further, the prior in period $n_g(p)+1$ is $f^{h(n_g(p))}(p)$, so that after $n_g(p)$ periods, information is disclosed by splitting beliefs between $1/2$ and 0. If $n_g(p) = \infty$, then information is never disclosed, so the second and third terms above equal 0 in the limit. ■

In the notation of Section 6.1, $n_g(f^h(1/2)) = n_g^h$ and $n_g(p_B^\ell) = n_g^\ell$.[37] Let $n_g(1/2) \coloneqq n_g^{1/2}$. Then:

**Claim 3.** The value of $\gamma'$ is a function of $n_g^\ell$ and $n_g^{1/2}$, where $n_g^\ell \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ and $n_g^{1/2} \in \{1, \infty\}$:

$$\gamma' = 2(1-\delta)\frac{(1-\delta^{n_g^\ell+1})U^h(1/2, n_g^{1/2}) - \delta(1-\delta^{n_g^{1/2}})U^h(p_B^\ell, n_g^\ell)}{\beta_g(n_g^\ell, n_g^{1/2})} \tag{B.4}$$

where $\beta_g(n_g^\ell, n_g^{1/2}) \coloneqq (1-\delta^{n_g^\ell+1})(1-2\delta^{n_g^{1/2}}f^{h(n_g^{1/2})}(1/2)) + 2\delta^{n_g^\ell+1}(1-\delta^{n_g^{1/2}})f^{h(n_g^\ell)}(p_B^\ell) > 0$.

*Proof.* Using equation (B.3), the value of $\gamma'$ above is derived as the solution of a system of two equations in two unknowns, $\gamma(1/2)$ and $\gamma(0)$. From the proof of Lemma 7, observe that $n_g^{1/2} = \infty$ in Cases 1 and 3, while $n_g^{1/2} = 1$ in Cases 2 and 4. The value of $n_g^\ell$, on the other hand, depends on the "branch" of $\gamma$ on which $p_B^\ell$ lies. Finally, the denominator $\beta_g(n_g^\ell, n_g^{1/2})$ is the sum of a strictly positive and a non-negative term. ■

**Lemma 8.** The payoff from the greedy policy is concave if and only if $x \geq \bar{x}$.

*Proof.* By Lemma 7, it is sufficient to show that $\gamma' \geq m_\infty^h \iff x \geq \bar{x}$. For each $(n_g^\ell, n_g^{1/2})$ pair, $\gamma' - m_\infty^h = C(n_g^\ell, n_g^{1/2}) \cdot (x - \bar{x})$, where $C(n_g^\ell, n_g^{1/2}) = \frac{2(1-\delta)(1-\delta^{n_g^{1/2}})(1+\delta p_B^\ell - \delta p_G^h)}{\beta_g(n_g^\ell, n_g^{1/2})(1-\delta s^h)} > 0$. Therefore, the payoff from the greedy policy is concave if and only if $x \geq \bar{x}$. ■

From the above, it follows that when $x = \bar{x}$, $\gamma' = m_\infty^h$.

[37] Recall from Section 6.1 that $n_\pi^a$ measures the number of periods of the quiet spell that follows a disclosure that induces action $a$. Recall further that under the greedy policy, the prior following the high and low actions is $f^h(1/2)$ and $p_B^\ell$, respectively.

## B.2 One-Shot Deviation Condition

I now turn to the one-shot deviation condition. Since the Sender already discloses no information when $p \geq 1/2$, we only need to verify the condition for $p < 1/2$. Let $D(p; x, y) \coloneqq \gamma(p) - \{(1 - \delta)k^{\ell}(p) + \delta\gamma(f^{\ell}(p))\}$ $\forall p \in [0, 1/2)$.[38,39] The following lemmas provide some useful simplifications:

**Lemma 9.** For a given $x$, if the greedy policy is optimal for some $\hat{y}$, then it is optimal for all $y \leq \hat{y}$.

*Proof.* Fix a value of $x \in [0, 1]$ and suppose that the greedy policy is optimal when $y = \hat{y}$. From equation (B.3), it follows that the value of $y$ does not impact $\gamma$. Further, by assumption and by Lemma 1, $D(p; x, \hat{y}) \geq 0$ $\forall p \in [0, 1/2)$. The result then follows from the fact that $D(p; x, y)$ is decreasing in $y$. ■

**Lemma 10.** For a given $(x, y)$, the greedy policy is optimal if and only if $\gamma$ is concave in $p$ and $D'_{+}(0; x, y) \geq 0$.

*Proof.* Fix $(x, y) \in [0, 1]^2$. If the greedy policy is optimal, by Lemma 1, $\gamma$ is concave in $p$ and $D(p; x, y) \geq 0$ $\forall p \in [0, 1/2)$. Since $D(0; x, y) = \gamma(0) - \delta\gamma(f^{\ell}(0)) = 0$, $p = 0$ minimises $D(\cdot; x, y)$ over this interval. Hence, $D'_{+}(0; x, y) \geq 0$.

Conversely, suppose $\gamma$ is concave in $p$ and $D'_{+}(0; x, y) \geq 0$. This implies that $\gamma(f^{\ell}(\cdot))$ is concave as it is the composition of a concave and an affine function. Further, $\gamma$ is affine on $[0, 1/2)$, so $D(\cdot; x, y)$ is convex on this interval. Since $D(0; x, y) = 0$ and $D'_{+}(0; x, y) \geq 0$, it follows that $D(p; x, y) \geq pD'_{+}(0; x, y) \geq 0$ for every $p \in [0, 1/2)$, so that by Lemma 1, the greedy policy is optimal. ■

**Claim 4.** It suffices to evaluate $D'_{+}(0; x, y)$ at the extreme values $x = \bar{x}$ and $x = 1$.

*Proof.* For fixed transition probabilities, $y$ and $p$, by equation (B.3), $\gamma(p)$ is affine in $x$, so that $D'_{+}(0; x, y)$ is also affine in $x$. By Lemma 8, $\gamma$ is concave if and only if $x \geq \bar{x}$, so the relevant extreme values are $\bar{x}$ and 1. ■

[38] Note that $D$ depends on $x$ and $y$ via functions $\gamma$ and $k^{\ell}$, respectively.
[39] By continuity of $\gamma$, this also gives the condition at $p = 1/2$.

As with $\gamma'$, the value of $D'_+(0;x,y)$ depends on the branch of $\gamma$ to which $p_B^\ell$ belongs.[40] For $n_g^\ell \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $D'_+(0;x,y)$ is given by:

$$D'_+(0;x,y) = \gamma' - (1-\delta)y - \delta s^\ell \left\{ m^h_{n_g^\ell} + (\delta s^h)^{n_g^\ell} \gamma' \right\}$$

so that:

$$D'_+(0;\bar{x},y) = (1-\delta)(\bar{y}-y) \geq 0 \iff y \leq \bar{y} \tag{B.5}$$

and

$$D'_+(0;1,y) = \gamma'(1 - \delta s^\ell (\delta s^h)^{n_g^\ell}) - (1-\delta)y. \tag{B.6}$$

where $\gamma'$ is as defined in equation (B.4). Since $D'_+(0;\cdot,\bar{y})$ is affine in $x$ and vanishes at $x = \bar{x}$ by equation (B.5), we have that:

$$D'_+(0;1,\bar{y}) = (1-\bar{x}) \cdot \frac{\partial D'_+(0;x,\bar{y})}{\partial x} \tag{B.7}$$

## B.3 Optimality

Summarising the previous two subsections, by Lemma 8, the payoff from the greedy policy is concave if and only if $x \geq \bar{x}$. By Lemma 9, Lemma 10 and Claim 4, if $D'_+(0;\bar{x},\hat{y}) \geq 0$ and $D'_+(0;1,\hat{y}) \geq 0$, then the greedy policy is optimal for all $(x,y) \in [0,1]^2$ with $x \geq \bar{x}$ and $y \leq \hat{y}$.

**Proposition 6.** Suppose $s_G \geq 0$. Then the greedy policy is optimal if and only if $x \geq \bar{x}$.

*Proof.* If $s_G \geq 0$, then $\bar{y} \geq 1$. By Lemma 8, the greedy policy is not optimal if $x < \bar{x}$. By the discussion above with $\hat{y} = 1$, it is optimal for all $x \geq \bar{x}$ and $y \in [0,1]$ if $D'_+(0;\bar{x},1) \geq 0$ and $D'_+(0;1,1) \geq 0$. By equation (B.5), $D'_+(0;\bar{x},1) = (1-\delta)(\bar{y}-1) \geq 0$. It only remains to show that $D'_+(0;1,1) \geq 0$.

Suppose first that $n_g^{1/2} = \infty$. From the proof of Lemma 7, this implies $n_g^\ell \in \{0,1,\infty\}$. Using equations (B.4) and (B.6) to evaluate $D'_+(0;1,1)$, we get:

$$D'_+(0;1,1) = \frac{1-\delta}{\beta_g(n_g^\ell,\infty)} \left\{ 1 + \delta^{n_g^\ell+1} - 2\delta^{n_g^\ell+1} f^{h(n_g^\ell)}(p_G^\ell) \right\}$$

[40] If $p_B^\ell$ is a kink of $\gamma$, read $n_g^\ell$ in the display below and in equations (B.6)–(B.8) as $n_g(f^\ell(\varepsilon))$ for small $\varepsilon > 0$; equation (B.4) holds with either value, since $\gamma$ is continuous.

and the term in the parentheses is strictly positive $\forall n_g^\ell \in \{0, 1, \infty\}$.

Suppose instead that $n_g^{1/2} = 1$, i.e. $f^h(1/2) < 1/2$. From the proof of Lemma 7, this implies that $n_g^\ell \in \mathbb{Z}_{\geq 0}$:

(a) If $n_g^\ell = 0$, then $D'_+(0; 1, 1) = \frac{(1-\delta)^2}{\beta_g(0,1)} \left\{1 - \delta p_G^\ell + \delta p_B^h + \delta s_G\right\} > 0$ since $s_G \geq 0$ and $1 - \delta p_G^\ell > 0$.

(b) If $n_g^\ell = 1$, then $D'_+(0; 1, 1) = \frac{(1-\delta)^2}{\beta_g(1,1)} \left\{(1-\delta)(1 + 2\delta f^h(1/2)) + 2\delta^2[p_G^h(1 - p_G^\ell) + p_B^h p_G^\ell]\right\}$, so that the term in the curly brackets is strictly positive.

(c) If $n_g^\ell \geq 2$, then $s^h > 0$,[41] and:

$$\begin{aligned} D'_+(0; 1, 1) &= \frac{(1-\delta)^2}{\beta_g(n_g^\ell, 1)} \left\{1 + \delta^{n_g^\ell+1}(1 - 2f^{h(n_g^\ell)}(p_G^\ell)) - \delta(1 - 2f^h(1/2)) \sum_{i=0}^{n_g^\ell} \delta^i\right\} \\ &\geq \frac{(1-\delta)^2}{\beta_g(n_g^\ell, 1)} \left\{1 - \delta^{n_g^\ell+1} v_{n_g^\ell+1}^h - \sum_{i=0}^{n_g^\ell} \delta^{i+1}(s^h v_i^h - v_{i+1}^h)\right\} \\ &= \frac{(1-\delta)^2}{\beta_g(n_g^\ell, 1)} \left\{(1 - \delta s^h) \sum_{i=0}^{n_g^\ell} \delta^i v_i^h\right\} \end{aligned}$$

where $v_i^h \coloneqq 2f^{h(i)}(1) - 1 \in [-1, 1]$. The above uses the facts that (i) since $s^h > 0$ and $s_G \geq 0$, $f^{h(n_g^\ell)}(p_G^\ell) \leq f^{h(n_g^\ell)}(p_G^h) = f^{h(n_g^\ell+1)}(1)$ and (ii) $1 - 2f^h(1/2) = s^h v_i^h - v_{i+1}^h$.

**Claim 5.** The term $\sum_{i=0}^{n_g^\ell} \delta^i v_i^h$ is bounded below by $1 - \delta^{n_g^\ell}$.

*Proof.* The claim is trivially true when $n_g^\ell \leq 1$. If $n_g^\ell \geq 2$, then $s^h > 0$, and recall that $n_g^\ell = \min\{i : f^{h(i)}(p_B^\ell) < 1/2\}$. Thus, for $i \in \{0, 1, \ldots, n_g^\ell - 1\}$, $f^{h(i)}(p_B^\ell) \geq 1/2$ and $f^{h(n_g^\ell)}(p_B^\ell) < 1/2$. It follows that for $i \in \{0, 1, \ldots, n_g^\ell - 1\}$, $f^{h(i)}(1) \geq f^{h(i)}(p_B^\ell) \geq 1/2$. Thus $v_i^h \geq 0$ for $i \in \{0, 1, \ldots, n_g^\ell - 1\}$ with $v_0^h = 1$, so the only term in $\sum_{i=0}^{n_g^\ell} \delta^i v_i^h$ that could be negative is $\delta^{n_g^\ell} v_{n_g^\ell}^h$, which is bounded below by $-\delta^{n_g^\ell}$. Therefore, $\sum_{i=0}^{n_g^\ell} \delta^i v_i^h \geq 1 + \delta^{n_g^\ell} v_{n_g^\ell}^h \geq 1 - \delta^{n_g^\ell}$. Note that the proof did not use the fact that $s_G \geq 0$. ■

Thus, for $n_g^\ell \geq 2$, $\sum_{i=0}^{n_g^\ell} \delta^i v_i^h$ is bounded below by $1 - \delta^{n_g^\ell} > 0$, so $D'_+(0; 1, 1) > 0$ and both conditions of Lemma 1 are satisfied. ■

[41] If $s^h \leq 0$, then for all $p \geq 1/2$, $f^h(p) \leq f^h(1/2) < 1/2$, so $n_g^\ell \leq 1$.

Proposition 6 provides a characterisation of the optimal policy when $s_G \geq 0$. What happens when $s_G < 0$?

Observe that $D'_+(0; x, y)$ is affine in both $x$ and $y$, with all second-order partial derivatives equalling 0. Therefore, for every $x \geq \bar{x}$, the value of $y$ satisfying $D'_+(0; x, y) = 0$ gives the maximum value of $y$ for which the greedy policy is optimal. Moreover, $D'_+(0; x, y) = 0$ defines $y$ as an affine function of $x$. Given $x$, let $L_g(x)$ denote the value of $y$ for which $D'_+(0; x, y) = 0$. By equation (B.5), $D'_+(0; \bar{x}, \bar{y}) = (1-\delta)(\bar{y} - \bar{y}) = 0$, so $L_g(\bar{x}) = \bar{y}$. Therefore, $L_g(x) = \bar{y} + m_g(x - \bar{x})$, where $m_g$ is the slope of $L_g(\cdot)$.

Define $Y$ to be the value of $y$ such that $D'_+(0; 1, Y) = 0$. Thus, $m_g = (Y - \bar{y})/(1 - \bar{x})$. By equations (B.6) and (B.7), $(1-\delta)m_g = \frac{\partial D'_+(0;x,\bar{y})}{\partial x}$. Define:

$$\Xi_{n_g^\ell} := \begin{cases} \sum_{i=0}^{n_g^\ell} \delta^i v_i^h, & \text{if } n_g^{1/2} = 1 \\ \dfrac{1 - \delta^{n_g^\ell + 1} v^h_{n_g^\ell + 1}}{1 - \delta s^h}, & \text{if } n_g^{1/2} = \infty \end{cases}$$

Using equation (B.4), it is then straightforward to show that:[42]

$$m_g = \frac{(1 - \delta^{n_g^{1/2}})}{\beta_g(n_g^\ell, n_g^{1/2})} \left\{ (1 - \delta s^\ell)\, \Xi_{n_g^\ell} + 2\delta(-s_G) \sum_{i=0}^{n_g^\ell - 1} (\delta s^h)^i \right\}. \tag{B.8}$$

The lemma below follows from the above discussion and the fact that $y \leq 1$:

**Lemma 11.** The greedy policy is optimal if and only if $x \geq \bar{x}$ and $y \leq \min\{1, L_g(x)\}$.

## Appendix C The Judicious Policy

Recall that the judicious policy maximises the probability with which the low action is taken by the Receiver each period. Formally:

$$(\alpha_j(p), q_j^h(p)) = \begin{cases} (2p - 1, 1), & p > 1/2 \\ (0, p), & p \leq 1/2 \end{cases}$$

Observe that the judicious policy completely reverses the greedy policy: the Sender reveals no

[42] This decomposition follows from the fact that $f^{h(i)}(p_G^\ell) = f^{h(i)}(p_G^h) + (-s_G)(s^h)^i = f^{h(i+1)}(1) + (-s_G)(s^h)^i$.

information when $p \leq 1/2$, and splits beliefs between $1/2$ and 1 when $p > 1/2$. Let $\Lambda(\cdot)$ denote the Sender's long-run payoff from using the judicious policy:

$$\Lambda(p) = \begin{cases} 2(1-p)\Lambda(1/2) + (2p-1)\Lambda(1), & p > 1/2 \\ (1-\delta)k^\ell(p) + \delta\Lambda(f^\ell(p)), & p \leq 1/2 \end{cases} \tag{C.1}$$

When $p = 1$, the Receiver is certain that the state is good, so he chooses the high action. This implies that the next period's prior is $f^h(1) = p_G^h$, so that $\Lambda(1) = (1-\delta) + \delta\Lambda(p_G^h)$. Moreover, for $p > 1/2$, $\Lambda(\cdot)$ is affine, with a slope of $\Lambda' \coloneqq \Lambda'_+(1/2) = 2[\Lambda(1) - \Lambda(1/2)]$.

Since the judicious policy is a complete reversal of the greedy policy, the arguments for concavity of $\Lambda(\cdot)$ and the one-shot deviation condition are symmetric to those for the greedy policy. Therefore, the judicious policy equivalents of Lemmas 7–10 are stated in the following subsections without proof.

## C.1 Concavity

The first lemma is a condition on the slope $\Lambda'$:

**Lemma 12.** The payoff from the judicious policy is concave if and only if $\Lambda' \leq m_\infty^\ell$.

Briefly, the proof rests on showing that $\Lambda(\cdot)$ is piecewise affine in each of four cases, depending on the range of values of $f^\ell([0, 1/2])$.[43] These are illustrated in Figure 7. It is then sufficient to show that $\Lambda$ is concave at the kinks, i.e. $\Lambda'_-(p) \geq \Lambda'_+(p)$ at kink $p$; this reduces to the condition above.

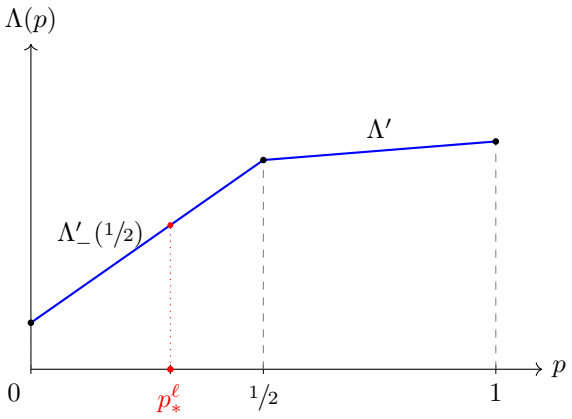
Let $n_j(p) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ be the number of periods in which no information is disclosed before the belief *rises* above $1/2$, i.e. $n_j(p) = \min\{i : f^{\ell(i)}(p) > 1/2\}$. Using equation (A.4), for $p \in [0, 1)$, the long-run payoff from the judicious policy (C.1) can be re-written as:

$$\Lambda(p) = (1-\delta)U^\ell(p, n_j(p)) + \delta^{n_j(p)}\left\{2(1 - f^{\ell(n_j(p))}(p))\Lambda(1/2) + (2f^{\ell(n_j(p))}(p) - 1)\Lambda(1)\right\} \tag{C.2}$$

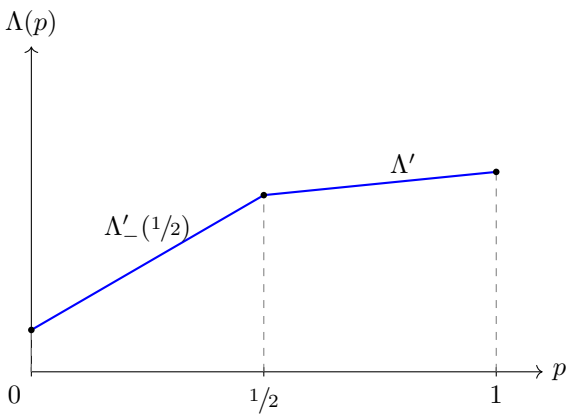
where $\Lambda(1) = (1-\delta) + \delta\Lambda(p_G^h)$. In the notation of Section 6.1, $n_j(f^\ell(1/2)) = n_j^\ell$ and $n_j(p_G^h) = n_j^h$.[44] Let $n_j^{1/2} \coloneqq n_j(1/2)$. Using equation (C.2), the value of $\Lambda'$ is a function of $n_j^{1/2}$ and $n_j^h$, where

[43] Under the judicious policy, the Receiver takes the low action only if $p \leq 1/2$, so the relevant range of beliefs following the low action is $f^\ell([0, 1/2])$.

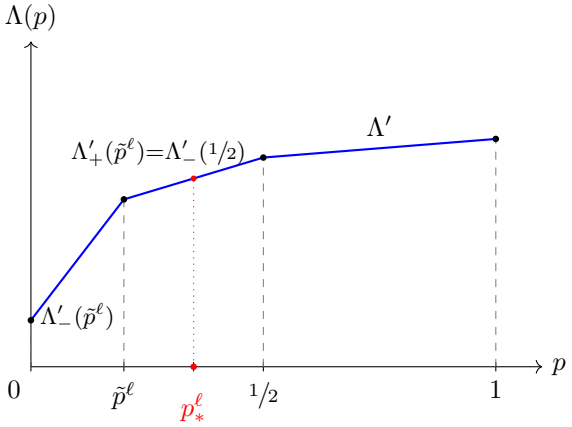
[44] Since the prior following a high action is $p_G^h$, $n_j^h$ measures the length of the quiet spell following the *high* action.



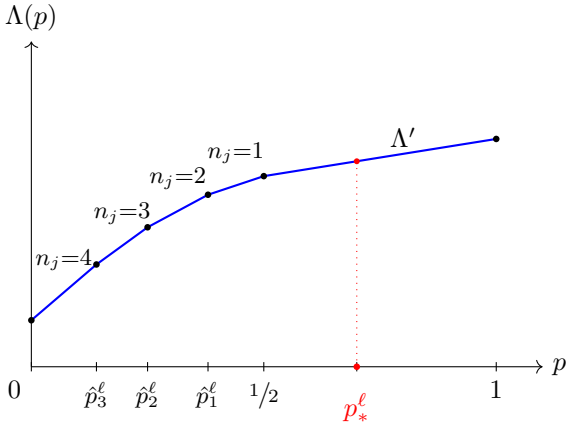

(a) Absorbing case: $f^\ell([0,1/2]) \subseteq [0,1/2]$




(b) Reverting case: $f^\ell([0,1/2]) \subseteq (1/2,1]$




(c) Partially absorbing case: $f^\ell(1/2) \leq 1/2 < f^\ell(0)$




(d) Cascading case: $f^\ell(1/2) > 1/2 \geq f^\ell(0)$

Figure 7: The Sender's payoff $\Lambda(\cdot)$ under the judicious policy of Section C.1; $p^\ell_*$ is the fixed point of $f^\ell(\cdot)$, drawn for $|s^\ell| < 1$

$n_j^{1/2} \in \{1, \infty\}$ and $n_j^h \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$:

$$\Lambda' = 2(1-\delta)\frac{(1-\delta^{n_j^{1/2}})[1+\delta U^\ell(p_G^h, n_j^h)] - (1-\delta^{n_j^h+1})U^\ell(1/2, n_j^{1/2})}{\beta_j(n_j^{1/2}, n_j^h)} \tag{C.3}$$

where $\beta_j(n_j^{1/2}, n_j^h) \coloneqq (1-\delta^{n_j^h+1})[1-2\delta^{n_j^{1/2}}(1-f^{\ell(n_j^{1/2})}(1/2))] + 2\delta^{n_j^h+1}(1-\delta^{n_j^{1/2}})[1-f^{\ell(n_j^h)}(p_G^h)] > 0$.

From the above, Lemma 12 reduces to a simple condition on the preference parameter $y$:

**Lemma 13.** The payoff from the judicious policy is concave if and only if $y \geq \bar{y}$.

As with the proof of Lemma 8, it turns out that $m_\infty^\ell - \Lambda' = \chi(n_j^{1/2}, n_j^h) \cdot (y - \bar{y})$ for each $(n_j^{1/2}, n_j^h)$ pair, where $\chi(n_j^{1/2}, n_j^h) = \frac{2(1-\delta)(1-\delta^{n_j^{1/2}})(1+\delta p_B^\ell - \delta p_G^h)}{\beta_j(n_j^{1/2}, n_j^h)(1-\delta s^\ell)} > 0$. In particular, when $y = \bar{y}$, then $\Lambda' = m_\infty^\ell$.

## C.2 One-Shot Deviation Condition

To show that the judicious policy is optimal, we now need to verify that the Sender does not strictly gain from a one-shot deviation to no disclosure, i.e. we must show that $\mathcal{D}(p; x, y) \coloneqq \Lambda(p) - \{(1-\delta)k^h(p) + \delta\Lambda(f^h(p))\} \geq 0 \ \forall p \in (1/2, 1]$.[45] The judicious policy equivalents of Lemmas 9 and 10 are as follows:

**Lemma 14.** For a given $y$, if the judicious policy is optimal for some $x = \hat{x}$, then it is optimal for all $x \in [0, \hat{x}]$.

**Lemma 15.** For a given $(x, y)$, the judicious policy is optimal if and only if $\Lambda(\cdot)$ is concave in $p$ and $\mathcal{D}'_-(1; x, y) \leq 0$.

Again, since $\mathcal{D}'_-(1; x, y)$ is affine in $y$, if $\bar{y} < 1$, it suffices to evaluate $\mathcal{D}'_-(1; x, y)$ at the extreme values $y = \bar{y}$ and $y = 1$. As with $\Lambda'$, the value of $\mathcal{D}'_-(1; x, y)$ depends on the branch of $\Lambda$ to which $p_G^h$ belongs.[46] For $n_j^h \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, $\mathcal{D}'_-(1; x, y)$ is given by:

$$\mathcal{D}'_-(1; x, y) = \Lambda' - (1-\delta)(1-x) - \delta s^h \left\{ m_{n_j^h}^\ell + (\delta s^\ell)^{n_j^h} \Lambda' \right\}$$

[45] Continuity of $\Lambda$ also gives the condition at $p = 1/2$.

[46] If $p_G^h$ is a kink of $\Lambda$, read $n_j^h$ in the display below and in equation (C.6) as $n_j(f^h(1-\varepsilon))$ for small $\varepsilon > 0$; equation (C.3) holds with either value, since $\Lambda$ is continuous.

where $\Lambda'$ is as defined in equation (C.3). Thus,

$$\mathcal{D}'_-(1; x, \bar{y}) = (1-\delta)(x - \bar{x}) \leq 0 \iff x \leq \bar{x} \tag{C.4}$$

Finally, since $\mathcal{D}'_-(1; \bar{x}, \cdot)$ is affine in $y$ and vanishes at $y = \bar{y}$ by equation (C.4), we have that:

$$\mathcal{D}'_-(1; \bar{x}, 1) = (1 - \bar{y}) \cdot \frac{\partial \mathcal{D}'_-(1; \bar{x}, y)}{\partial y} \tag{C.5}$$

## C.3 Optimality

By Lemma 13, the payoff from the judicious policy is concave if and only if $y \geq \bar{y}$. By Lemmas 14 and 15 and equations (C.4) and (C.5), an argument symmetric to that for the greedy policy shows that $\mathcal{D}'_-(1; x, y) = 0$ defines $x$ as an affine function of $y$ and given $y$, gives the maximum value of $x$ for which the judicious policy is optimal. We can analogously define the line $L_j(y) = \bar{x} + m_j(y - \bar{y})$ where $(1-\delta)m_j = -\frac{\partial \mathcal{D}'_-(1; \bar{x}, y)}{\partial y}$.

Let $v_i^\ell := 1 - 2f^{\ell(i)}(0) \in [-1, 1]$ denote the judicious policy counterpart of $v_i^h$, and define:

$$\Theta_{n_j^h} := \begin{cases} \sum_{i=0}^{n_j^h} \delta^i v_i^\ell, & \text{if } n_j^{1/2} = 1 \\ \dfrac{1 - \delta^{n_j^h+1} v^\ell_{n_j^h+1}}{1 - \delta s^\ell}, & \text{if } n_j^{1/2} = \infty \end{cases}$$

Using equation (C.3), it is then straightforward to show that:[47]

$$m_j = \frac{(1 - \delta^{n_j^{1/2}})}{\beta_j(n_j^{1/2}, n_j^h)} \left\{ (1 - \delta s^h)\, \Theta_{n_j^h} + 2\delta(-s_B) \sum_{i=0}^{n_j^h - 1} (\delta s^\ell)^i \right\}. \tag{C.6}$$

Thus, since $x \leq 1$, we have that:

**Lemma 16.** The judicious policy is optimal if and only if $y \geq \bar{y}$ and $x \leq \min\{1, L_j(y)\}$.

[47] This decomposition follows from the fact that: (i) $f^{\ell(i)}(p_B^h) = f^{\ell(i+1)}(0) - (-s_B)(s^\ell)^i$, and (ii) $2f^\ell(1/2) - 1 = s^\ell v_i^\ell - v_{i+1}^\ell$.

# Appendix D Full Disclosure

Let $\theta(\cdot)$ denote the payoff to the Sender from fully revealing the state to the Receiver, i.e. $(\alpha_{fd}(p), q^h_{fd}(p)) = (p, 1)$, $\forall p \in [0,1]$. For $p \in [0,1]$, the Sender's long-run expected payoff is given by:

$$\begin{aligned}\theta(p) &= (1-\delta)p + \delta p \cdot \theta(f^h(1)) + \delta(1-p) \cdot \theta(f^\ell(0)) \\ &= p \cdot \theta(1) + (1-p) \cdot \theta(0)\end{aligned}$$

where $\theta(0) = \delta\theta(p^\ell_B)$ and $\theta(1) = (1-\delta) + \delta\theta(p^h_G)$. Thus, $\theta(\cdot)$ is affine over $[0,1]$. Since the above is true for all $p \in [0,1]$, it holds at $f^h(1) = p^h_G$ and $f^\ell(0) = p^\ell_B$ as well. It is then easy to solve for the Sender's long-run payoff when she fully discloses the state every period:

$$\theta(p) = \frac{(1-\delta)p + \delta p^\ell_B}{1 + \delta p^\ell_B - \delta p^h_G}$$

**Proposition 7.** Full disclosure is optimal if and only if $x \leq \bar{x}$ and $y \leq \bar{y}$.

*Proof.* Since $\theta(\cdot)$ is affine and hence concave, using Lemma 1, it only remains to show that the Sender gets a weakly higher payoff than a one-shot deviation to providing no information. Thus, if $p \geq 1/2$, we require that:

$$\theta(p) \geq (1-\delta)(p + (1-p)x) + \delta\theta(f^h(p)) \iff x \leq \bar{x}$$

and if $p \leq 1/2$, we require that:

$$\theta(p) \geq (1-\delta)py + \delta\theta(f^\ell(p)) \iff y \leq \bar{y}$$ ■

# Appendix E Omitted Proofs

## E.1 Proof of Proposition 1

If states evolve exogenously, $s_G = 0$ and $s_B = 0$. Hence, $\bar{x} = 0$ and $\bar{y} = 1$. The result follows from Proposition 6. ■

## E.2 Proof of Proposition 2

For all possible values of the transition probabilities, it is always the case that $\bar{x} \leq 1$ and $\bar{y} \geq 0$. Thus, when $(x, y) = (1, 0)$, by Lemma 8, the payoff induced by the greedy policy is concave.

Further, when $x = 1$, the numerator of $\gamma'$ in equation (B.4) is $2(1-\delta)(1-\delta^{n_g^{1/2}}) > 0$. Thus, by Lemma 10 and equation (B.6), $D'_+(0; 1, 0) = \gamma'(1 - \delta s^\ell (\delta s^h)^{n_g^\ell}) > 0$, proving the result. ■

## E.3 Proof of Theorem 1

Under positive externalities, $s_G > 0$ and $s_B > 0$. Hence, $\bar{x} \leq 0$ and $\bar{y} \geq 1$. The result then follows from Proposition 6. ■

## E.4 Proof of Theorem 2

Under negative externalities, $s_G < 0$ and $s_B < 0$. Hence, $\bar{x} \geq 0$ and $\bar{y} \leq 1$.

**Part (i):** restates Proposition 7.

**Part (ii):** The statement of the result restates Lemma 11. It only remains to show that $m_g > 0$, so that $L_g$ is strictly increasing.

In equation (B.8), we only need to show that the term in the parentheses is positive. Under negative externalities, $-s_G > 0$, so the second term is non-negative. For the first term, since $|\delta s^a| < 1$, it suffices to show that $\Xi_{n_g^\ell}$ is positive. If $n_g^{1/2} = \infty$, this is immediate: $v^h_{n_g^\ell + 1} \leq 1$, so that $\Xi_{n_g^\ell} \geq (1 - \delta^{n_g^\ell + 1})/(1 - \delta s^h) > 0$.

If $n_g^{1/2} = 1$, i.e. $f^h(1/2) < 1/2$, then it suffices to show that $\sum_{i=0}^{n_g^\ell} \delta^i v_i^h$ is positive. The sum equals 1 when $n_g^\ell = 0$. It follows from Claim 5 that the sum is bounded below by $1 - \delta^{n_g^\ell} > 0$ when $n_g^\ell \geq 1$. Thus, $m_g > 0$.

**Part (iii):** The statement of the result restates Lemma 16. It only remains to show that $m_j > 0$.

Again, in equation (C.6), we only need to show that the term in the parentheses is positive. Under negative externalities, $-s_B > 0$, so the second term is non-negative. For the first term, since $|\delta s^a| < 1$, it only remains to show that $\Theta_{n_j^h}$ is positive. If $n_j^{1/2} = \infty$, this is immediate: $v^\ell_{n_j^h + 1} \leq 1$,

so that $\Theta_{n_j^h} \geq (1-\delta^{n_j^h+1})/(1-\delta s^\ell) > 0$. If $n_j^{1/2} = 1$, i.e. $f^\ell(1/2) > 1/2$, this implies $n_j^h \in \mathbb{Z}_{\geq 0}$ and we have the following:

**Claim 6.** The term $\sum_{i=0}^{n_j^h} \delta^i v_i^\ell$ is bounded below by $1-\delta^{n_j^h}$.

*Proof.* The claim is trivially true if $n_j^h \leq 1$. If $n_j^h \geq 2$, $s^\ell > 0$,[48] and recall that $n_j^h = \min\{i : f^{\ell(i)}(p_G^h) > 1/2\}$. Thus, for $i \in \{0, 1, \ldots, n_j^h - 1\}$, $f^{\ell(i)}(0) \leq f^{\ell(i)}(p_G^h) \leq 1/2$ and $f^{\ell(n_j^h)}(p_G^h) > 1/2$. An argument symmetric to Claim 5 proves the result. ■

Since the sum equals 1 when $n_j^h = 0$ and is at least $1-\delta^{n_j^h} > 0$ otherwise, we have shown that $m_j > 0$.

**Part (iv):** Parts (i)–(iii) give necessary and sufficient conditions for the optimality of the greedy, judicious and full disclosure policies, and $m_g, m_j > 0$. Let $\mathcal{R} \coloneqq \{(x,y) \in [0,1]^2 : x > \bar{x}, y > \bar{y}, y > L_g(x), x > L_j(y)\}$ be the set of $(x,y)$ for which none of these three policies are optimal.

**Non-emptiness of $\mathcal{R}$.** If $\mathcal{R}$ is empty, then Parts (i)-(iii) cover the whole unit square, and there is nothing left to prove. The following result characterises when $\mathcal{R}$ is non-empty.

**Lemma 17.** $m_g m_j \leq 1$, and $\mathcal{R}$ is non-empty if and only if $m_g m_j < 1$.

*Proof.* By parts (ii) and (iii), $m_g > 0$ and $m_j > 0$. So, for $x > \bar{x}$, the conditions $y > L_g(x)$ and $x > L_j(y)$ together read $m_g(x-\bar{x}) < y - \bar{y} < (x-\bar{x})/m_j$ and they imply $y > \bar{y}$. If $(x,y) \in \mathcal{R}$, dividing by $x - \bar{x} > 0$ gives $m_g < 1/m_j$. Conversely, if $m_g m_j < 1$, pick $k \in (m_g, 1/m_j)$. Then $(\bar{x}+\varepsilon, \bar{y}+k\varepsilon)$ satisfies the inequalities above for every $\varepsilon > 0$, and it lies in $[0,1]^2$ for $\varepsilon$ small, because $\bar{x}, \bar{y} \in (0,1)$ when $\delta > 0$; if $\delta = 0$, then $m_g = m_j = 1$.

It remains to show that $m_g m_j \leq 1$. Suppose instead that $m_g m_j > 1$, and pick $k \in (1/m_j, m_g)$. For $\varepsilon > 0$ small, $(x,y) \coloneqq (\bar{x}+\varepsilon, \bar{y}+k\varepsilon)$ lies in $[0,1]^2$ and satisfies $x > \bar{x}$, $y > \bar{y}$, $y < L_g(x)$ and $x < L_j(y)$. By parts (ii) and (iii), both the greedy and the judicious policies are optimal at $(x,y)$. Since the judicious policy discloses nothing at priors below $1/2$, $V = w^\ell$ on $[0, 1/2)$. Since the greedy policy is optimal, $V = \gamma$. Hence $D(\cdot; x, y) = 0$ on $[0, 1/2)$, so $D'_+(0; x, y) = 0$, i.e. $y = L_g(x)$. This contradicts $y < L_g(x)$. ■

[48] If $s^\ell \leq 0$, then for all $p \leq 1/2$, $f^\ell(p) \geq f^\ell(1/2) > 1/2$, so $n_j^h \leq 1$.

Equivalently, the following result characterises when $\mathcal{R}$ is non-empty in terms of the transition probabilities:

**Lemma 18.** Suppose $\delta > 0$. Then $\mathcal{R}$ is empty if and only if neither $f^\ell([0, 1/2])$ nor $f^h([1/2, 1])$ contains beliefs on both sides of $1/2$.

*Proof.* I say that $f^\ell$ *crosses* $1/2$ if $f^\ell([0, 1/2])$ contains beliefs both above and below $1/2$, and that $f^h$ crosses $1/2$ if $f^h([1/2, 1])$ does. A policy whose split attains the maximum in (3.5) at every prior is optimal, since $V$ is then the fixed point of the policy's own Bellman operator, a contraction.

( $\Longrightarrow$ ): If $\mathcal{R}$ is empty, by Lemma 17, $L_g = L_j$. Since $\bar{x}, \bar{y} \in (0, 1)$ when $\delta > 0$, this line contains points $(x, y) \in [0, 1]^2$ with $x > \bar{x}$, at which both the greedy and the judicious policies are optimal by parts (ii) and (iii) above. There, $V$ is affine on each half, $V = w^\ell$ on $[0, 1/2]$ and $V = w^h$ on $[1/2, 1]$. If $f^\ell$ crosses $1/2$, then $V \circ f^\ell = (w^\ell - (1-\delta)k^\ell)/\delta$ is affine on $[0, 1/2]$, although $f^\ell$ maps $[0, 1/2]$ onto beliefs on both sides of $1/2$; so $V$ has no kink at $1/2$. The same holds if $f^h$ crosses $1/2$. Then $V$ is affine on $[0, 1]$, and full disclosure attains the maximum at every prior, so it is optimal although $x > \bar{x}$. This contradicts part (i) of Theorem 2, proved above.

( $\Longleftarrow$ ): Let $W$ be affine on $[0, 1/2]$ and on $[1/2, 1]$. Since $f^\ell([0, 1/2])$ and $f^h([1/2, 1])$ each lie in one half, $w^\ell_W$ is affine on $[0, 1/2]$ and $w^h_W$ on $[1/2, 1]$ (with $w_W$ as in the proof of Claim 1). So $w_W$ lies below the piecewise-linear interpolation of its values at 0, $1/2$ and 1, and $TW$ is the concave hull of these three points, which is again affine on each half. As $T^n 0 \to V$ uniformly (Lemma 3), $V$ is affine on each half, and so are $w^\ell$ on $[0, 1/2]$ and $w^h$ on $[1/2, 1]$. Recall that $V \geq w$, with equality at 0 and 1. If $V(1/2) = w^h(1/2)$, then $V - w^h$ is affine and non-negative on $[1/2, 1]$ and zero at both ends, so $V = w^h$ there; since $V$ is affine on $[0, 1/2]$, the greedy policy attains the maximum at every prior. If $V(1/2) = w^\ell(1/2)$, the judicious policy does, symmetrically. Otherwise $V(1/2) > w(1/2)$, so $V$ is affine around $1/2$ (Lemma 5), hence on $[0, 1]$, and full disclosure attains the maximum at every prior. So full disclosure, the greedy or the judicious policy is optimal at every $(x, y)$, and $\mathcal{R}$ is empty. ■

Assume from now on that $m_g m_j < 1$. Let $\Gamma_g := \{(x, L_g(x)) : x > \bar{x}\} \cap [0, 1]^2$ and $\Gamma_j := \{(L_j(y), y) : y > \bar{y}\} \cap [0, 1]^2$ denote the two edges of $\mathcal{R}$ that start at $(\bar{x}, \bar{y})$. Both are segments of

positive length; the greedy policy is optimal on $\Gamma_g$ and the judicious policy is optimal on $\Gamma_j$.

Recall from Appendix A.2 that $w^a(q) = (1-\delta)k^a(q) + \delta V(f^a(q))$ denotes the Sender's payoff from inducing the posterior $q$ together with action $a \in \{h, \ell\}$ and then continuing optimally. By Corollary 4, for all $(x, y) \in \mathcal{R}$, $V(1/2) = \max\{w^h(1/2), w^\ell(1/2)\}$. Let $\mathcal{V}(x, y) \coloneqq w^h(1/2) - w^\ell(1/2)$ be the Sender's gain from inducing action $h$ rather than action $\ell$ at $q = 1/2$. Fix $(x, y) \in \mathcal{R}$.

**The no-disclosure line.** If $\mathcal{V}(x, y) = 0$, then $w^h(1/2) = w^\ell(1/2)$ and by Lemma 5 and Corollary 4, any interval in $I$ would have $1/2$ as an endpoint with $w^h(1/2) \neq w^\ell(1/2)$. So $I = \emptyset$, i.e. $V = w$, and disclosing no information at any prior is optimal. Let $N(\cdot)$ denote the Sender's payoff from never disclosing information, with the Receiver taking $h$ at the belief $1/2$. Along the no-disclosure path from any prior the sequence of actions is fixed, so $N(p)$ is affine in $(x, y)$ for each $p$. Let

$$\mathcal{N}(x, y) \coloneqq \big[(1-\delta)k^h(1/2) + \delta N\big(f^h(1/2)\big)\big] - \big[(1-\delta)k^\ell(1/2) + \delta N\big(f^\ell(1/2)\big)\big],$$

which is affine in $(x, y)$. I refer to $\Gamma_N \coloneqq \{(x, y) : \mathcal{N}(x, y) = 0\}$ as the *no-disclosure line*. Two observations link it to the optimal policy.

(a) *If disclosing no information is optimal at $(x, y)$, then $\mathcal{N}(x, y) = \mathcal{V}(x, y) = 0$.* Indeed, optimality of no disclosure implies $V = w$. Since $V$, $w^h$ and $w^\ell$ are continuous at $1/2$, $w^\ell(1/2) = \lim_{p \uparrow 1/2} V(p) = V(1/2) = \lim_{p \downarrow 1/2} V(p) = w^h(1/2)$, so $\mathcal{V}(x, y) = 0$. Hence $V = w$ satisfies the recursion that defines $N$ (with action $h$ at $1/2$), so $V = N$, and therefore $\mathcal{N}(x, y) = \mathcal{V}(x, y) = 0$.

(b) *If the greedy policy is optimal at $(x, y)$, then $\mathcal{V}(x, y) \geq 0$, and if $\mathcal{V}(x, y) = 0$ then disclosing no information is also optimal.* Indeed, by (B.1), $V(1/2) = \gamma(1/2) = w^h(1/2)$, while $V(1/2) \geq w^\ell(1/2)$ by (A.5). If, in addition, $\mathcal{V}(x, y) = 0$, then $V(1/2) = w(1/2)$ and $w^h(1/2) = w^\ell(1/2)$, so by Lemma 5, $I = \emptyset$. Symmetrically, using (C.1), if the judicious policy is optimal at $(x, y)$ then $\mathcal{V}(x, y) \leq 0$, and if $\mathcal{V}(x, y) = 0$ then disclosing no information is also optimal.

In particular, at $(\bar{x}, \bar{y})$, both the greedy and the judicious policies are optimal, so $V = w$ on $[1/2, 1]$ (greedy) and on $[0, 1/2)$ (judicious). Disclosing no information is therefore optimal at $(\bar{x}, \bar{y})$, and by (a), the no-disclosure line passes through $(\bar{x}, \bar{y})$.

**Truncated policies.** If $\mathcal{V}(x,y) > 0$, then $V(1/2) \geq w^h(1/2) > w^\ell(1/2)$. By continuity of $V$ and $w^\ell$ at $1/2$, $V(p) > w^\ell(p) = w(p)$ for all $p < 1/2$ close enough to $1/2$, so $I \neq \emptyset$. By Lemma 5 and Corollary 4, $I = (\underline{p}, 1/2)$ with $\underline{p} < 1/2$. If $\underline{p} = 0$, the greedy policy would be optimal, which is impossible on $\mathcal{R}$. So $\underline{p} \in (0, 1/2)$, and the truncated greedy policy with $q = \underline{p}$ is optimal. If $\mathcal{V}(x,y) < 0$, the symmetric argument shows that the truncated judicious policy with some $q \in (1/2, 1)$ is optimal.

Now, for each $p$, $V(p)$ is continuous in $(x,y)$: changing $(x,y)$ by at most $\varepsilon$ in each coordinate changes the payoff of every policy by at most $\varepsilon$. Hence $\mathcal{V}$ is continuous on $[0,1]^2$ and $\mathcal{V} \neq 0$ on $\mathcal{R}$ off the no-disclosure line. There are two cases.

*Case A: the no-disclosure line $\Gamma_N$ intersects $\mathcal{R}$.* Since it passes through $(\bar{x}, \bar{y})$, its slope lies strictly between $m_g$ and $1/m_j$, and it divides $\mathcal{R}$ into two non-empty convex (hence connected) sets: $\mathcal{R}_+$, between $\Gamma_g$ and $\Gamma_N$, and $\mathcal{R}_-$, between $\Gamma_N$ and $\Gamma_j$. $\mathcal{V}$ has a constant sign on each.

Suppose $\mathcal{V} < 0$ on $\mathcal{R}_+$. Since every point of $\Gamma_g$ close to $(\bar{x}, \bar{y})$ is a limit of points in $\mathcal{R}_+$, continuity gives $\mathcal{V} \leq 0$ there, and by point (b) above $\mathcal{V} \geq 0$, so $\mathcal{V} = 0$ and disclosing no information is optimal at these points. This implies that by point (a) above, $\mathcal{N} = 0$ on $L_g$ starting at $(\bar{x}, \bar{y})$, so the no-disclosure line is $L_g$, which does not intersect $\mathcal{R}$, a contradiction. Hence $\mathcal{V} > 0$ on $\mathcal{R}_+$, where a truncated greedy policy is optimal. Symmetrically, $\mathcal{V} < 0$ on $\mathcal{R}_-$, where a truncated judicious policy is optimal. Finally, at points of $\mathcal{R}$ on the line, $\mathcal{V} = 0$ by continuity, so disclosing no information is optimal by the previous discussion on the no-disclosure line.

*Case B: the no-disclosure line does not intersect $\mathcal{R}$.* Then $\mathcal{V} \neq 0$ on the connected set $\mathcal{R}$, so it has a constant sign. If $\mathcal{V} > 0$ on $\mathcal{R}$, the argument of Case A applied to $\Gamma_j$ shows that disclosing no information is optimal on $\Gamma_j$ and that the no-disclosure line coincides with $L_j$. Then all of $\mathcal{R}$ lies below the no-disclosure line, and a truncated greedy policy is optimal throughout. If $\mathcal{V} < 0$ on $\mathcal{R}$, the no-disclosure line coincides with $L_g$, all of $\mathcal{R}$ lies above it, and a truncated judicious policy is optimal throughout.

In either case, the no-disclosure line passes through $(\bar{x}, \bar{y})$ and has slope in $[m_g, 1/m_j]$, so it is strictly increasing. The optimal policy is truncated greedy below it and truncated judicious above it, and disclosing no information is optimal on it.

**Characterising the truncated threshold.** It only remains to characterise $\underline{q}$ and $\overline{q}$. Recall from Section 6.1 that, under the truncated greedy policy, $I_{tg} = (\underline{q}, 1/2)$, $F_{tg} = f^\ell$ on $[0, \underline{q}]$ and $F_{tg} = f^h$ on $[1/2, 1]$. Under the truncated judicious policy, $I_{tj} = (1/2, \overline{q})$, $F_{tj} = f^\ell$ on $[0, 1/2]$ and $F_{tj} = f^h$ on $[\overline{q}, 1]$.

**Lemma 19.** Suppose $I = (\underline{q}, 1/2)$ for some $\underline{q} \in (0, 1/2)$, so that the truncated greedy policy with threshold $\underline{q}$ is optimal. Then exactly one of the following holds:

(i) $F_{tg}^{(n)}(z_{tg}^\ell) = 1/2$ for some $n \geq 0$, and $F_{tg}^{(i)}(z_{tg}^\ell) \notin [\underline{q}, 1/2]$ for $i < n$;

(ii) the path $F_{tg}^{(n)}(z_{tg}^\ell)$ never meets $[\underline{q}, 1/2]$.

Symmetrically, suppose $I = (1/2, \overline{q})$ for some $\overline{q} \in (1/2, 1)$ so that the truncated judicious policy with threshold $\overline{q}$ is optimal. Then either $F_{tj}^{(n)}(z_{tj}^h) = 1/2$ for some $n \geq 0$ and $F_{tj}^{(i)}(z_{tj}^h) \notin [1/2, \overline{q}]$ for $i < n$, or the path from $z_{tj}^h$ never meets $[1/2, \overline{q}]$.

*Proof.* Consider the truncated greedy policy; the judicious case is symmetric. By Lemma 5, $I = (\underline{q}, 1/2)$, $V$ is affine on $[\underline{q}, 1/2]$ with slope $\rho$ say, and $V > w = w^\ell$ on $I$. Further, $V = w^\ell$ on $[0, \underline{q}]$, and $V = w^h$ on $[1/2, 1]$, because $V(1/2) = w^h(1/2)$. Recall that $z_{tg}^\ell = f^\ell(\underline{q})$.

We first show that $V$ is not affine on any neighbourhood of $z_{tg}^\ell$. If it were, then $w^\ell(p) = (1-\delta)k^\ell(p) + \delta V(f^\ell(p))$ would be affine on an open interval around $\underline{q}$. Its slope there equals $V'_-(\underline{q}) \geq V'_+(\underline{q}) = \rho$, because $w^\ell = V$ to the left of $\underline{q}$, and is at most $\rho$, because $w^\ell \leq V$ to the right of $\underline{q}$ with equality at $\underline{q}$. Hence $w^\ell = V$ just to the right of $\underline{q}$, contradicting $V > w = w^\ell$ on $I$.

Now suppose that neither (i) nor (ii) holds. Then the path from $z_{tg}^\ell$ first meets $[\underline{q}, 1/2]$ at some time $n$, at a point of $[\underline{q}, 1/2)$. Before time $n$ it lies in $(0, \underline{q})$, where $V = w^\ell$, or in $(1/2, 1)$, where $V = w^h$. As in the proof of Lemma 6, for a closed interval in either set, the slope drop of $V$ across it equals $\delta|s^a|$ times the slope drop of $V$ across its image under $f^a = F_{tg}$. For $\zeta > 0$, let $\psi(\zeta)$ be the slope drop of $V$ across $[z_{tg}^\ell - \zeta, z_{tg}^\ell + \zeta]$. Applying this $n$ times, with $\zeta$ small, gives

$$\psi(\zeta) \leq \Big(\delta \max_{a \in \mathcal{A}} |s^a|\Big)^n \, [V'_-(c) - V'_+(d)],$$

where $[c, d]$ is an interval of length at most $2\zeta$ centred at the point where the path meets $[\underline{q}, 1/2)$. If that point lies in $I$, then $[c, d] \subseteq I$ for $\zeta$ small, and the bracket is zero. If it is $\underline{q}$, then $[c, d] =$

$[\underline{q}-\zeta', \underline{q}+\zeta']$ with $\zeta' \leq \zeta$. Since $\underline{q}+\zeta' \in I$, $V = w^\ell$ on $[0, \underline{q}]$, and $V'_+(\underline{q}) \geq (w^\ell)'_+(\underline{q})$ as in the proof of Lemma 6,

$$\begin{aligned} V'_-(c) - V'_+(d) &\leq (w^\ell)'_-(\underline{q}-\zeta') - (w^\ell)'_+(\underline{q}) \\ &= \delta|s^\ell|\,[\text{slope drop of } V \text{ across } f^\ell([\underline{q}-\zeta', \underline{q}])] \\ &\leq \delta \max_{a\in\mathcal{A}} |s^a| \psi(\zeta), \end{aligned}$$

because $f^\ell([\underline{q}-\zeta', \underline{q}])$ contains $z^\ell_{tg}$ and has length at most $\zeta$. In either case, $\psi(\zeta) \leq \delta \max_{a\in\mathcal{A}} |s^a| \psi(\zeta)$, so $\psi(\zeta) = 0$ and $V$ is affine on $[z^\ell_{tg}-\zeta, z^\ell_{tg}+\zeta]$, a contradiction. ■

Lemma 19 characterises the thresholds. Suppose (i) holds, and let $a_1, \ldots, a_n$ be the actions taken along the path from $z^\ell_{tg}$ to $1/2$: $a_i = \ell$ if $F^{(i-1)}_{tg}(z^\ell_{tg}) < \underline{q}$ and $a_i = h$ if $F^{(i-1)}_{tg}(z^\ell_{tg}) \geq 1/2$. Then $\underline{q}$ solves the affine equation

$$f^{a_n}(\cdots f^{a_1}(f^\ell(\underline{q}))) = \frac{1}{2} \tag{E.1}$$

So $\underline{q}$ belongs to a countable set of candidate thresholds that depends only on the transition probabilities, and not on $\delta$, $x$ or $y$; the preference parameters only select among the candidates. The simplest candidate, with $n = 0$, is $\underline{q} = (1/2 - p^\ell_B)/s^\ell$, which lies in $(0, 1/2)$ if and only if $1/2$ lies strictly between $p^\ell_B$ and $f^\ell(1/2)$. In Example 3, this gives $\underline{q} = (0.5-0.4)/0.5 = 0.2$. The same holds for $\overline{q}$, with the path from $z^h_{tj} = f^h(\overline{q})$ and $h$ in place of $\ell$ in (E.1).

In case (ii) of Lemma 19, $n^\ell_{tg} = \infty$: after a disclosure that induces $\ell$, the Sender never discloses again.

This completes the proof. ■

## E.5 Proof of Proposition 3

**Part (i):** Differentiating,

$$\frac{\partial \bar{x}}{\partial \delta} = \frac{-s_B}{(1+\delta p^\ell_B - \delta p^h_G)^2} > 0, \qquad \frac{\partial \bar{y}}{\partial \delta} = \frac{s_G}{(1+\delta p^\ell_B - \delta p^h_G)^2} < 0.$$

**Part (ii):** The probability $p^h_B$ only appears in the numerator of $\bar{x}$, so $\partial \bar{x}/\partial p^h_B = -\delta/(1+\delta p^\ell_B - \delta p^h_G) < 0$ and does not affect $\bar{y}$. Symmetrically, $p^\ell_G$ only appears in the numerator of $\bar{y}$, so $\partial \bar{y}/\partial p^\ell_G =$

$-\delta/(1+\delta p_B^\ell - \delta p_G^h) < 0$ and does not affect $\bar{x}$.

**Part (iii):** Differentiating,

$$\frac{\partial \bar{x}}{\partial p_G^h} = \frac{-\delta^2 s_B}{(1+\delta p_B^\ell - \delta p_G^h)^2}, \qquad \frac{\partial \bar{y}}{\partial p_G^h} = \frac{\delta[1-\delta s^\ell]}{(1+\delta p_B^\ell - \delta p_G^h)^2},$$

$$\frac{\partial \bar{x}}{\partial p_B^\ell} = \frac{\delta(1-\delta s^h)}{(1+\delta p_B^\ell - \delta p_G^h)^2}, \qquad \frac{\partial \bar{y}}{\partial p_B^\ell} = \frac{-\delta^2 s_G}{(1+\delta p_B^\ell - \delta p_G^h)^2},$$

and by assumption, $\delta > 0$, $\delta s^a < 1$, $s_G < 0$ and $s_B < 0$, so all four expressions above are strictly positive. ■

## E.6 Proof of Lemma 2

Recall that $\tau = \min\{t : p_t \in I_\pi\}$ is the first period in which the Sender provides information.

**Part (i):** Before $\tau$ the prior lies outside $I_\pi$, where the Sender discloses nothing and the belief moves by $F_\pi$.

**Part (ii):** In any period $t$ with $p_t \in I_\pi$, the posterior is $q_\pi^h$ with probability $\alpha_\pi(p_t)$ and $q_\pi^\ell$ otherwise, so $p_{t+1} \in \{z_\pi^h, z_\pi^\ell\}$. Starting from $z_\pi^a$, the belief moves by $F_\pi$ through $F_\pi^{(n)}(z_\pi^a), n < n_\pi^a$, all outside $I_\pi$, until it reaches $r_\pi^a \in I_\pi$. Again the same argument applies. Induction on $t$ proves the result.

**Part (iii):** $P_\pi^a$ has at most $n_\pi^a + 1$ elements. In $P_\pi$, transitions from beliefs other than $r_\pi^h$ and $r_\pi^\ell$ are deterministic; transitions from $r_\pi^h$ and $r_\pi^\ell$ are as stated in (ii). ■

## E.7 Proof of Proposition 4

By equation (A.3), $p_G^h + p_B^h \geq 1 \iff f^h(1/2) = (p_G^h + p_B^h)/2 \geq 1/2 \iff p_*^h \geq 1/2$. By equation (A.2), $f^{h(n)}(1/2) = p_*^h - (s^h)^n(p_*^h - 1/2) \geq p_*^h - (p_*^h - 1/2) = 1/2$. So under the greedy policy, once the posterior $1/2$ is realised, the belief never re-enters $I_g = (0, 1/2)$ and it converges to $p_*^h$; that is $n_g^{1/2} = \infty$.

Recall that $\tau = \min\{t : p_t \in I_\pi\}$ is the first period in which the Sender provides information. If $\tau = \infty$, the belief never falls below $1/2$, the Receiver always takes the high action, and $p_t \to p_*^h$ by equation (A.2). Suppose instead $\tau < \infty$. Each disclosure induces the posterior $1/2$ with probability

$2p$, where $p$ is the current prior, after which disclosure stops. Otherwise it induces the posterior 0, after which the belief is $p_B^\ell$. If $n_g^\ell = \infty$, disclosure stops. If $n_g^\ell < \infty$, the next disclosure occurs at prior $r_g^\ell = f^{h(n_g^\ell)}(p_B^\ell) \in I_g$. This point is the same for every episode, so after the first disclosure, each subsequent one ends disclosure for good with the same probability $2r_g^\ell > 0$. Thus, the number of disclosures is finite almost surely.

A mirror-image argument applies for the judicious policy. ■

## E.8 Proof of Proposition 5

It suffices to prove Part (i); Part (ii) follows from a mirror-image argument. By equation (A.3), $p_G^h + p_B^h < 1 \iff p_*^h < 1/2$. By equation (A.2), since $f^{h(i)}(p) \to p_*^h$ for every $p$, $f^{h(i)}(p) < 1/2$ for some finite $i$, i.e. $n_g(p) < \infty$. In particular, $n_g^\ell < \infty$, and from any prior $p_1$, the first disclosure occurs in period $\tau = 1 + n_g(p_1) < \infty$. Consider any period $t$ in which the Sender discloses:

(i) If the disclosure induces action $h$, the posterior is $1/2$. The following period's prior is $f^h(1/2) < 1/2$ by assumption, so the Sender discloses again in period $t+1$.

(ii) If the disclosure induces action $\ell$, the posterior is 0 and $p_{t+1} = p_B^\ell$. By definition of $n_g^\ell$, $f^{h(i)}(p_B^\ell) \geq 1/2$ for $i \in \{0, 1, \ldots, n_g^\ell - 1\}$ and $f^{h(n_g^\ell)}(p_B^\ell) < 1/2$. So in periods $t+1, \ldots, t+n_g^\ell$ the Sender discloses nothing and the Receiver takes action $h$. The Sender discloses again in period $t + n_g^\ell + 1$.

In either case the next disclosure occurs after finitely many periods, and so the Sender discloses in infinitely many periods, on every path. ■

## E.9 Proof of Corollary 1

Under negative externalities, $(p_G^\ell + p_B^\ell) - (p_G^h + p_B^h) = -(s_G + s_B) > 0$. So $p_G^h + p_B^h \geq 1$ implies $p_G^\ell + p_B^\ell > 1$, and the claim follows from Propositions 4 and 5. Under full disclosure, $I_{fd} = (0, 1)$ and every prior after period 1 is in $\{p_B^\ell, p_G^h\} \subset (0, 1)$. ■

### E.10 Proof of Corollary 2

Consider the truncated greedy policy; the judicious case is symmetric. As in the proof of Theorem 2(iv), $I = I_{tg} = (\underline{q}, 1/2)$. Since $n^{\ell}_{tg} < \infty$, the path from $z^{\ell}_{tg}$ meets $I$, so case (ii) of Lemma 19 fails and (i) holds. So $F^{(n)}_{tg}(z^{\ell}_{tg}) = 1/2 \notin I_{tg}$ and $F^{(n+1)}_{tg}(z^{\ell}_{tg}) = f^h(1/2) = z^h_{tg}$. Hence $n^{\ell}_{tg} = n + 1 + n^h_{tg}$ and $r^{\ell}_{tg} = r^h_{tg}$. Every disclosure episode after the first therefore starts at $r_{tg} \coloneqq r^h_{tg}$, where action $h$ is induced with probability $\alpha_{tg}(r_{tg})$, independently of the past. ■

## Appendix F Mixed Externalities

By Definition 1, under positive (negative) externalities, the high action always improves (worsens) the future state compared to the low action, regardless of the current state. What happens when there are positive externalities in one state and negative externalities in the other, i.e. different actions improve the future state? There are two cases to consider: one where $s_G > 0$ and $s_B < 0$, and the other with the inequalities reversed.

**Proposition 8.** Suppose there are positive externalities in the good state and negative externalities in the bad state, i.e. $s_G > 0$ and $s_B < 0$. Then:

(i) Full disclosure is optimal if and only if $x \leq \bar{x}$,

(ii) The greedy policy is optimal if and only if $x \geq \bar{x}$.

*Proof.* Since $s_G > 0 > s_B$, we have that $\bar{x} \geq 0$ and $\bar{y} \geq 1$. Part (i) follows from Proposition 7. Part (ii) follows from Proposition 6. ■

The intuition for Proposition 8 is as follows. When $s_G > 0$ and $s_B < 0$, the action that matches the state is also the one that is better for the future: the high action keeps the good state persistent, and the low action makes the bad state more likely to turn good. In the good state the high action therefore dominates: it gives the Sender her highest stage payoff of $1 \geq y$ and also leads to a better future state. Therefore, if $\delta > 0$, the Sender never wants to induce the low action in the good state, which is why $\bar{y} > 1$, so that the judicious policy is never optimal. All that remains is how often to

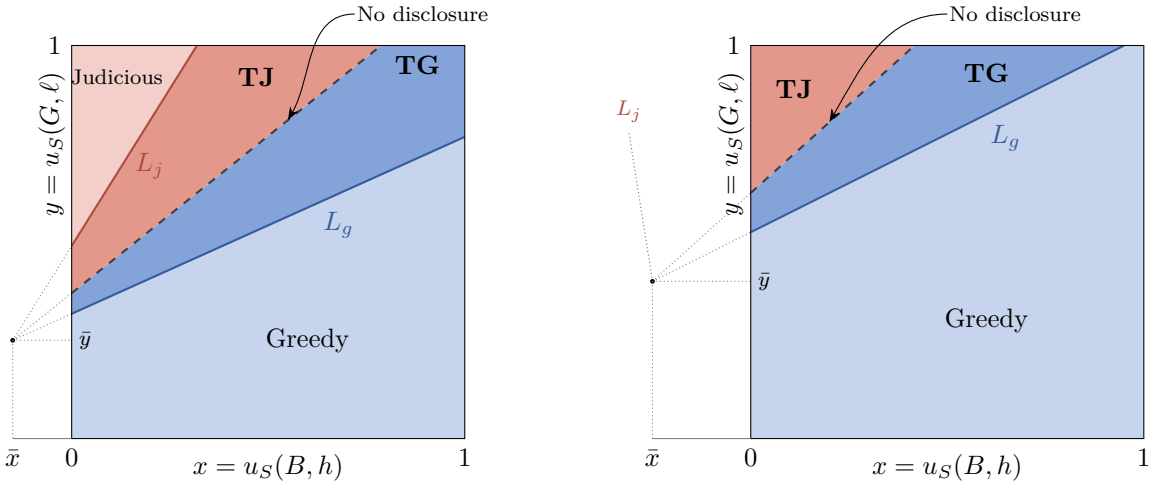
induce the high action in the bad state. Each such mistake gives the Sender $x$ today but makes the bad state more persistent. Because this trade-off is the same in every period, the solution is bang-bang. If $x \leq \bar{x}$, the current gain does not cover the discounted future loss, and the Sender fully discloses so that the Receiver never makes this mistake. If $x \geq \bar{x}$, she induces the high action as often as possible, which is the greedy policy.

**Proposition 9.** Suppose there are negative externalities in the good state and positive externalities in the bad state, i.e. $s_G < 0$ and $s_B > 0$. Then:

(i) The greedy policy is optimal if and only if $y \leq \min\{1, L_g(x)\}$,

(ii) The judicious policy is optimal if and only if $y \geq \bar{y}$ and $x \leq \min\{1, L_j(y)\}$,

(iii) Otherwise, there exists a no-disclosure line, with the optimal policy being truncated greedy below it and truncated judicious above it.




(a) With a judicious region

(b) Without a judicious region

Figure 8: Optimal policies when $s_G < 0 < s_B$;
TG: truncated greedy, TJ: truncated judicious

*Proof.* Since $s_G < 0$ and $s_B > 0$, we have that $\bar{x} \leq 0$ and $\bar{y} \leq 1$, with strict inequalities if $\delta > 0$. By Proposition 7, full disclosure is not optimal for $x > 0$ (and for any $x$ if $\delta > 0$). Part (i) restates Lemma 11; in particular, since $s_G < 0$, $m_g > 0$ as in Theorem 2 (ii). Part (ii) restates Lemma 16.

However, since $s_B > 0$ now, it is not necessary that $m_j > 0$. If $m_j \leq 0$, then for any $(x, y) \in [0, 1]^2$, the judicious policy is never optimal (see Figure 8).

For part (iii), we apply the same argument as case (iv) of the proof of Theorem 2 on $\mathbb{R}^2$, as each result it uses holds for all $(x, y) \in \mathbb{R}^2$ (for Claim 1, see footnote 34). On $\mathbb{R}^2$ the edge $\Gamma_j$, along which the judicious policy is optimal, always exists, even when it lies outside the unit square (see Figure 8b). If $m_j \leq 0$, two steps of that argument change. First, $x > L_j(y)$ holds automatically when $x > \bar{x}$ and $y > \bar{y}$, so $\mathcal{R} = \{(x, y) : x > \bar{x}, y > \bar{y}, y > L_g(x)\}$ is non-empty and Lemma 17 is not needed. Second, in Case A the no-disclosure line passes through $(\bar{x}, \bar{y})$ with direction strictly between those of $\Gamma_g$ and $\Gamma_j$, rather than with slope between $m_g$ and $1/m_j$. It therefore need not be increasing, which part (iii) does not claim. Restricting to $[0, 1]^2$ gives the result. ■

When $s_G < 0$ and $s_B > 0$, the opposite is true: in each state, the action that does not match the state is the one that improves the future. Taking the low action in the good state, or the high action in the bad state, raises the probability that the next period's state is good. So Receiver mistakes now pay off twice: once through the current payoff ($x$ or $y$), and again through a better future state. This is why when $\delta > 0$, full disclosure, which eliminates all mistakes, is never optimal here: $\bar{x} < 0$. For the same reason, the payoff from the greedy policy is always concave. The question is therefore not whether the Sender should induce mistakes but in which state. The greedy policy induces them in the bad state and the judicious policy in the good state. The greedy policy is optimal when the Sender's preference for the high action is strong relative to her preference for the good state, that is, when $x$ is high and $y$ is low. The judicious policy is optimal in the reverse case; unlike under negative externalities, however, the judicious region may be empty. The judicious policy induces the future-improving mistake in the good state, but it does so by taking the low action in the bad state, which is the worse action for the future there. In between, the Sender induces mistakes in both states by keeping the belief close to $1/2$ through the truncated policies.

# Appendix G General Model

In this appendix, I adopt the general model of Ely (2017), allowing for intertemporal externalities. The Sender's and Receiver's period-$t$ payoffs depend on a state of the world $\omega_t \in \Omega$ and the Receiver's action $a_t \in \mathcal{A}$ in that period, and are given by $u_S(\omega_t, a_t)$ and $u_R(\omega_t, a_t)$, respectively. The sets $\Omega$ and $\mathcal{A}$ are finite, with $|\Omega| \geq 2$ and $|\mathcal{A}| \geq 2$. Since the Receiver is short-lived, he plays myopic best responses that maximise his stage game payoff, while the Sender maximises her long-run discounted expected payoff.

As earlier, the state transitions depend on both the period-$t$ action and the period-$t$ state. Formally, the state transition probabilities are given by $\mathbb{P}(\omega' \mid \omega, a)$, such that $\sum_{\omega' \in \Omega} \mathbb{P}(\omega' \mid \omega, a) = 1$. This includes the following cases:

(i) Absorbing states: If, for a given state $\omega$, $\mathbb{P}(\omega \mid \omega, a) = 1 \ \forall a \in \mathcal{A} \iff \mathbb{P}(\omega' \mid \omega, a) = 0$ $\forall \omega' \neq \omega, \forall a \in \mathcal{A}$, then state $\omega$ is absorbing: once the game reaches state $\omega$, it stays there forever. This is the case analysed in the basic model in Ely (2017).

(ii) Action-independent evolution: If, for every distinct pair $\omega, \omega' \in \Omega$, $\mathbb{P}(\omega' \mid \omega, a) = \mathbb{P}(\omega' \mid \omega, a')$, $\forall a, a' \in \mathcal{A}$, then the state transition process is independent of action, and only depends on the state. This is the case explored in Renault, Solan, and Vieille (2017),[49] the general model of Ely (2017), Ashkenazi-Golan et al. (2023),[50] and Lehrer and Shaiderman (2026).

(iii) State-independent evolution: If, for every $a \in \mathcal{A}$, $\mathbb{P}(\omega \mid \omega', a) = \mathbb{P}(\omega \mid \omega'', a)$, $\forall \omega$, $\omega'$, $\omega'' \in \Omega$, then the state transition process is independent of state, and only depends on the action.

The players' common prior distribution in period $t$ is represented by a vector $\mathbf{p_t} \in \Delta(\Omega)$. Let $p_t(\omega)$ denote the period-$t$ prior probability of being in state $\omega$, so that $\sum_{\omega \in \Omega} p_t(\omega) = 1$. The Sender chooses an experiment $(\Sigma, \sigma_t)$, defined in the same manner as earlier. Using the same arguments as in the basic model, it is straightforward to show that it is without loss of generality to assume that $|\Sigma| \leq |\mathcal{A}|$, with each signal realisation inducing a different action. The players update to their

[49] Renault, Solan, and Vieille (2017) analyse a special class of games in which the action space is binary.
[50] Ashkenazi-Golan et al. (2023) analyse a special class of games in which the state space is binary.

posterior belief $q_t(\omega) := \mathbb{P}(\omega_t = \omega \mid \varsigma_t)$ after observing the signal realisation $\varsigma_t \in \Sigma$ and the chosen experiment, using Bayes' rule.

After the Receiver chooses an action, the state evolves based on the transition probabilities described above. Given the posterior $\mathbf{q_t}$ in period $t$ and the action $a \in \mathcal{A}$ taken in period $t$, the prior in period $t+1$ is given by:

$$p_{t+1}(\omega') = \sum_{\omega \in \Omega} q_t(\omega) \cdot \mathbb{P}(\omega' \mid \omega, a) \tag{G.1}$$

Thus, the players' prior in a period is a conditional distribution over states given the signal realisations and actions in all periods preceding that period. Updating for every state $\omega \in \Omega$ using equation (G.1), we get the whole vector of period-$t+1$ beliefs. Given $\mathbf{q_t}$ and action $a \in \mathcal{A}$, let $f^a(\mathbf{q_t})$ denote period $t+1$'s prior.

Let the Receiver's best-response function be $a^*(\mathbf{q})$ and let $A^*(\mathbf{q})$ denote his set of best responses. I assume that the Receiver has an a.e.-unique best response. When $A^*(\mathbf{q})$ has more than one element, he takes the action in $A^*(\mathbf{q})$ that the Sender prefers, taking her continuation payoff into account. Thus, the Sender's interim expected payoffs in the stage game and the next period's prior from inducing a posterior $\mathbf{q_t}$ are given by $\mathbb{E}_{\mathbf{q_t}}(u_S(\mathbf{q_t})) := \mathbb{E}_{\mathbf{q_t}}(u_S(\omega_t, a))$ and $f^a(\mathbf{q_t})$, respectively, where $a \in A^*(\mathbf{q_t})$.

Let $V(\cdot)$ denote the Sender's maximum payoff from the dynamic optimisation problem. This function is characterised by the solution to the following Bellman equation:

$$\begin{gathered} V(\mathbf{p_t}) = \sup_{\mu \in \Delta(\Delta(\Omega))} \mathbb{E}_\mu \left\{ \max_{a \in A^*(\mathbf{q_t})} \left[ (1-\delta)\mathbb{E}_{\mathbf{q_t}}(u_S(\mathbf{q_t})) + \delta V(f^a(\mathbf{q_t})) \right] \right\} \\ \text{subject to } \mathbb{E}_\mu(\mathbf{q_t}) = \mathbf{p_t} \end{gathered} \tag{G.2}$$

In the functional equation above, the expression in the curly brackets is the Sender's interim expected payoff in the dynamic game. Concavification of these interim payoffs gives us the Sender's optimal expected payoff, so that:

**Theorem 3.** The functional equation (G.2) characterising the Sender's optimal expected payoff

has a unique fixed point and is given by:

$$V(\mathbf{p}) = \text{cav}\left\{\max_{a\in A^*(\mathbf{q})}\left[(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta V(f^a(\mathbf{q}))\right]\right\}.$$

*Proof.* The proof generalises that of Ely (2017). In order to prove the theorem, it is sufficient to show that the operator $TV(\mathbf{q}) = \text{cav}\left\{\max_{a\in A^*(\mathbf{q})}[(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta V(f^a(\mathbf{q}))]\right\}$ is a contraction mapping in the space of bounded real-valued functions on $\Delta(\Omega)$ with the sup norm. Blackwell's sufficient conditions[51] are satisfied as follows:

(i) Consider two functions $V(\cdot)$ and $W(\cdot)$. If $V(\mathbf{q}) \geq W(\mathbf{q})\ \forall \mathbf{q} \in \Delta(\Omega)$, then for every $\mathbf{q} \in \Delta(\Omega)$ and every $a \in A^*(\mathbf{q})$, $(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta V(f^a(\mathbf{q})) \geq (1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta W(f^a(\mathbf{q}))$. Taking the maximum over $a \in A^*(\mathbf{q})$ on both sides preserves the inequality. Since the concavification of a function is the smallest concave function greater than or equal to the said function, this implies that

$$\begin{aligned}&\text{cav}\left\{\max_{a\in A^*(\mathbf{q})}\left[(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta V(f^a(\mathbf{q}))\right]\right\}\\ \geq\ &\text{cav}\left\{\max_{a\in A^*(\mathbf{q})}\left[(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta W(f^a(\mathbf{q}))\right]\right\}, \forall \mathbf{q} \in \Delta(\Omega).\end{aligned}$$

Equivalently, $TV \geq TW$.

(ii) Consider a constant $c \geq 0$. Then, $(V+c)(\mathbf{q}) = V(\mathbf{q}) + c$ which implies that:

$$\begin{aligned}T(V+c)(\mathbf{q}) =\ &\text{cav}\left\{\max_{a\in A^*(\mathbf{q})}\left[(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta(V+c)(f^a(\mathbf{q}))\right]\right\}\\ =\ &\text{cav}\left\{\max_{a\in A^*(\mathbf{q})}\left[(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta V(f^a(\mathbf{q}))\right] + \delta c\right\}\\ =\ &\text{cav}\left\{\max_{a\in A^*(\mathbf{q})}\left[(1-\delta)\mathbb{E}_{\mathbf{q}}(u_S(\mathbf{q})) + \delta V(f^a(\mathbf{q}))\right]\right\} + \delta c\\ =\ &TV + \delta c,\end{aligned}$$

since adding the constant $\delta c$ to every term adds $\delta c$ to the maximum. Thus, the operator $T$ is a contraction (with Lipschitz constant $\delta < 1$, by assumption). By the contraction mapping theorem, the iterates of $T$ converge to its unique fixed point $V$. ■

[51] Theorem 3.3 of Stokey, Lucas, and Prescott (1989).

Ely (2017) proved a special case of the above theorem with $f(\mathbf{p})$ replacing $f^a(\mathbf{p})$. From the above, it follows that the Sender's value function is concave in the prior.

I use the above to generalise Lemma 1. The Sender's policy $(\mu(\mathbf{p}), \{\mathbf{q}^a(\mathbf{p})\}_{a\in\mathcal{A}})$ specifies the posteriors $\{\mathbf{q}^a\}_{a\in\mathcal{A}}$ and the distribution $\mu$ over them that the Sender would choose at each prior $\mathbf{p}$, such that $\mathbb{E}_\mu(\mathbf{q}^a) = \mathbf{p}$. Let $\phi(\mathbf{p})$ be the payoff to the Sender from following such a policy.

**Lemma 20.** A policy $(\mu(\mathbf{p}), \{\mathbf{q}^a(\mathbf{p})\}_{a\in\mathcal{A}})$ of the Sender is optimal if and only if $\phi(\cdot)$ is concave over $\Delta(\Omega)$ and the following is satisfied:

$$\phi(\mathbf{p}) \geq (1-\delta)\mathbb{E}_{\mathbf{p}}[u_S(\mathbf{p})] + \delta\phi(f^a(\mathbf{p})), \forall \mathbf{p} \in \Delta(\Omega), a \in A^*(\mathbf{p}) \tag{G.3}$$

The proof follows the same steps as the proof of Lemma 1 and is therefore omitted.